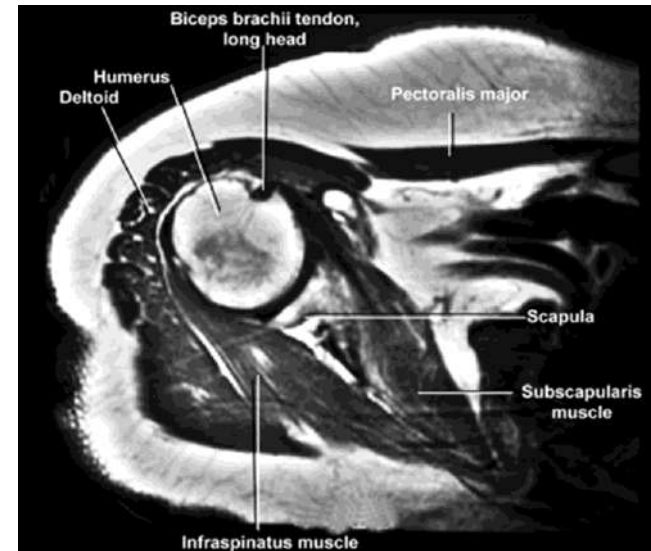
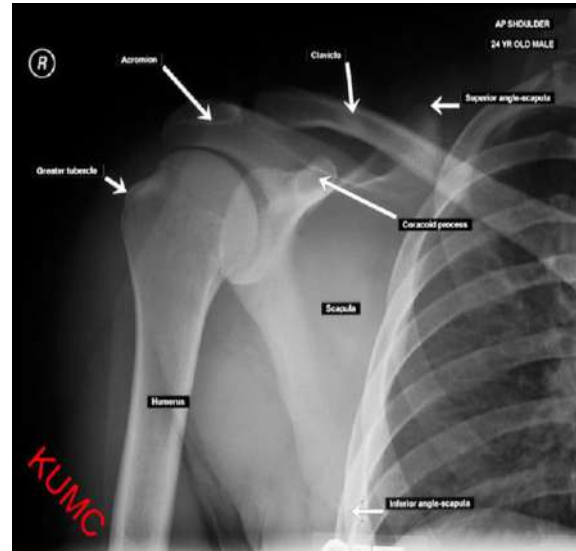
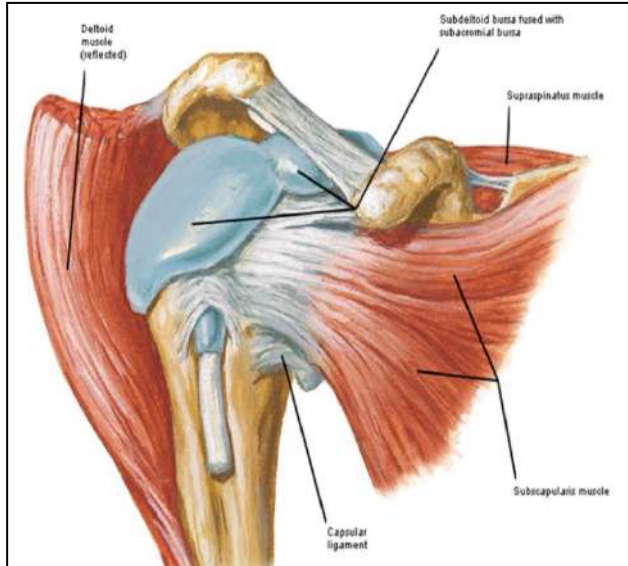


Joints of Upper Limb and Radiology



Prof. Laila M. Aboul Mahasen Morsy
Anatomy Dept. ,Faculty of Medicine , KAU

Objectives :

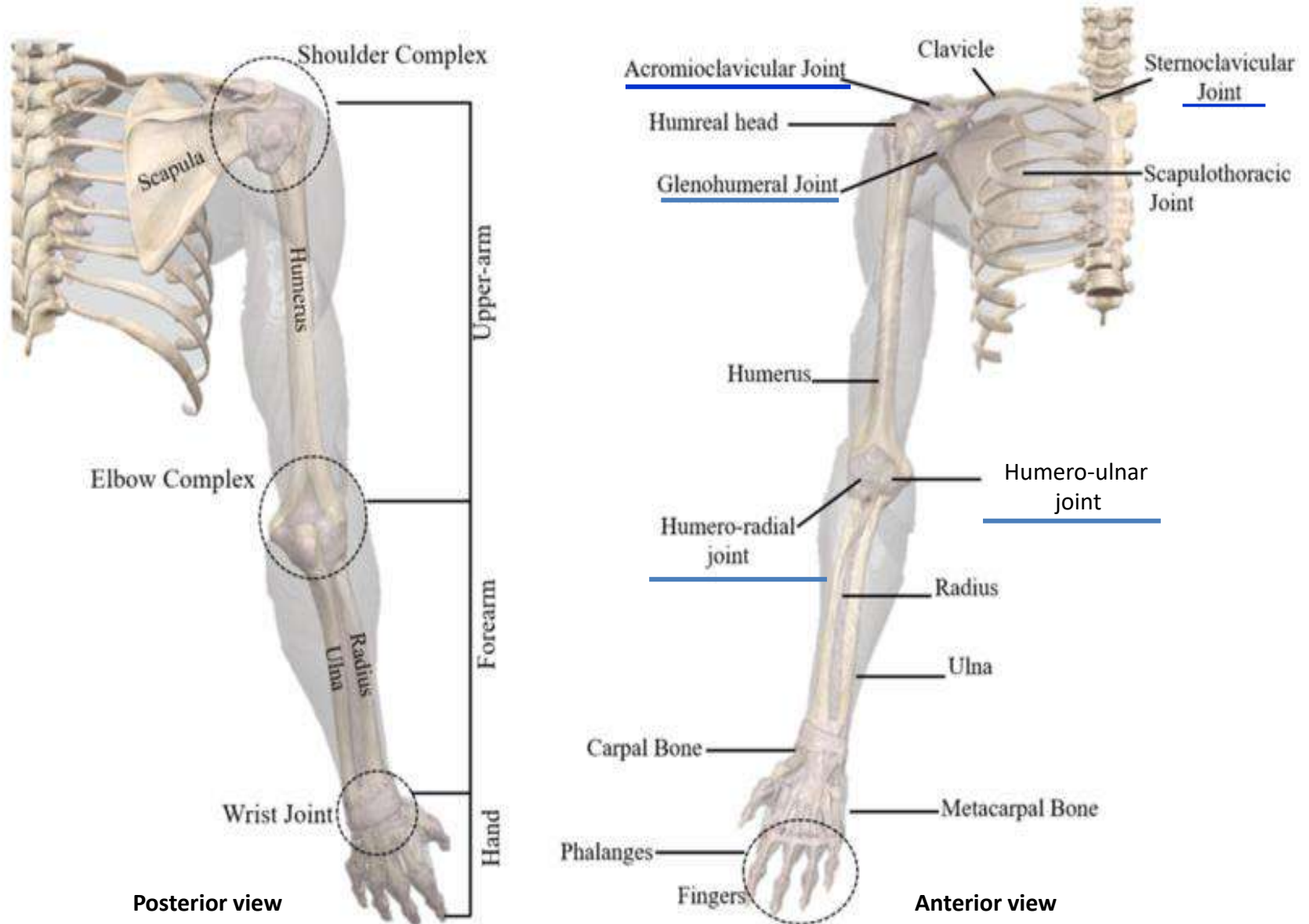
By the end of this lecture, you should be able to:

1. Define and classify joints of upper limb according to their anatomical structure, movement and shape of its articular surfaces.
2. Describe the main structures of **the shoulder, elbow and wrist joints** (articular surfaces, ligaments, synovial membrane, capsule ... etc).
3. List factors that maintain the stability of upper limb joints.
4. Enumerate different movements in the joints.
5. Define Hilton law (Nerve supply of muscles and joints).
6. Knowing the MSK anatomy on the plain x -radiographs.

***Clinical Note:**

1. Identifying fractures on **radiographs**.
2. Knowing the basic anatomy on shoulder **MRI**

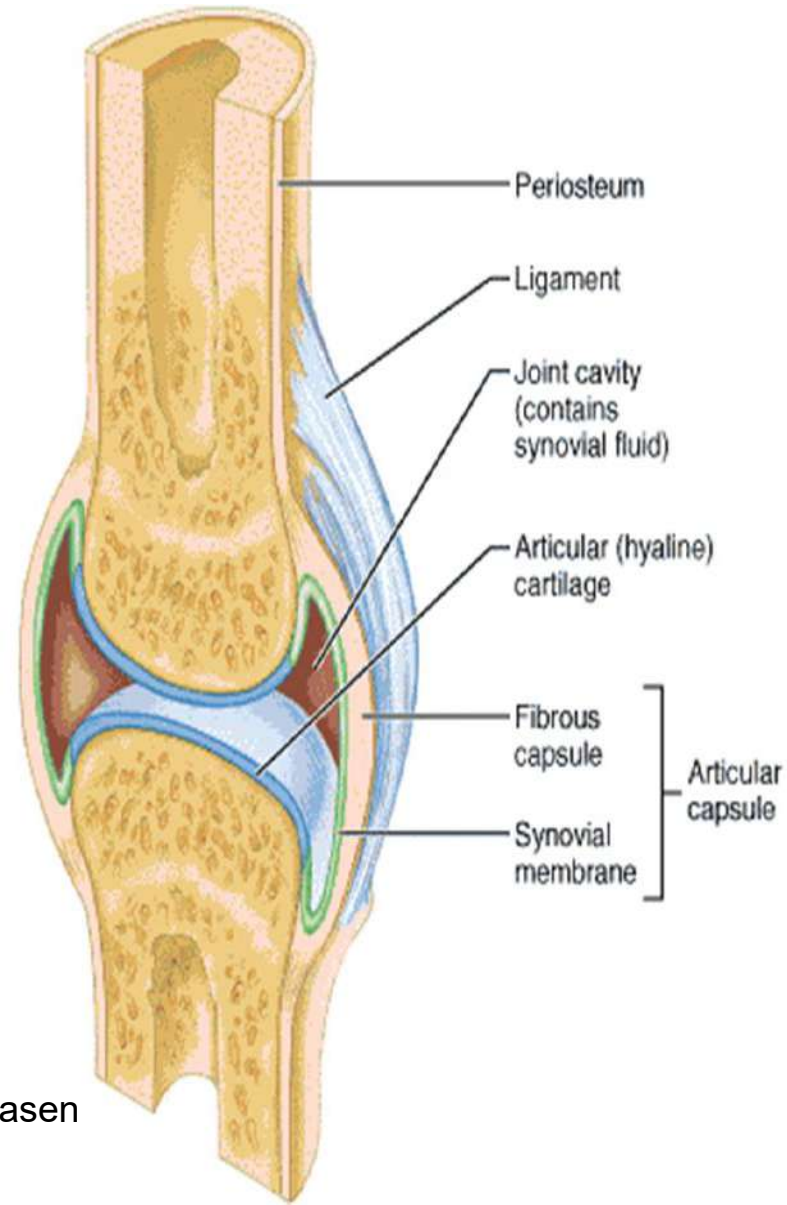
Joints of the Upper Limb



Remember!

General characters of SYNOVIAL JOINTS

- They are freely movable
- Structure of synovial joint :
 1. There is **a joint cavity**
 2. The articular surfaces are covered by **hyaline cartilage**
 3. The bones are held together by a **fibrous capsule**
 4. The capsule maybe thickened in certain parts to form **ligaments**
 5. The capsule is lined by a **synovial membrane &**
The synovial membrane secretes **synovial fluid**
- ❑ **The synovial membrane** does not cover the articular cartilage



Sternoclavicular joint

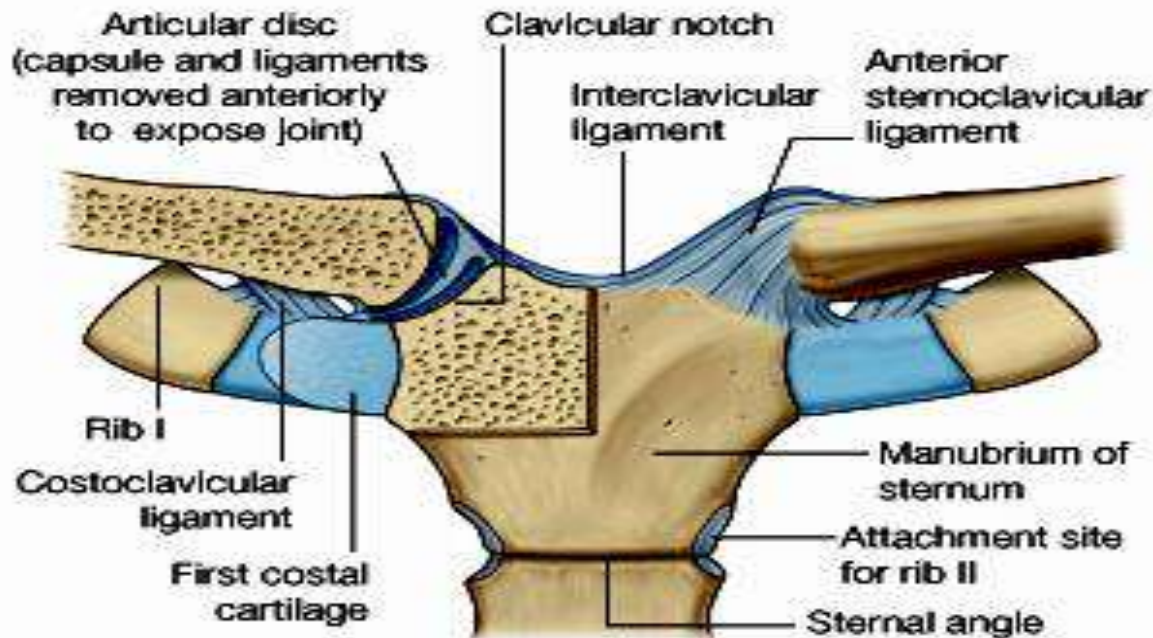
Type : saddle synovial joint

Articular surfaces:

1. Medial end of the clavicle
2. clavicular notch of manubrium sterni with a small part of 1st costal cartilage.

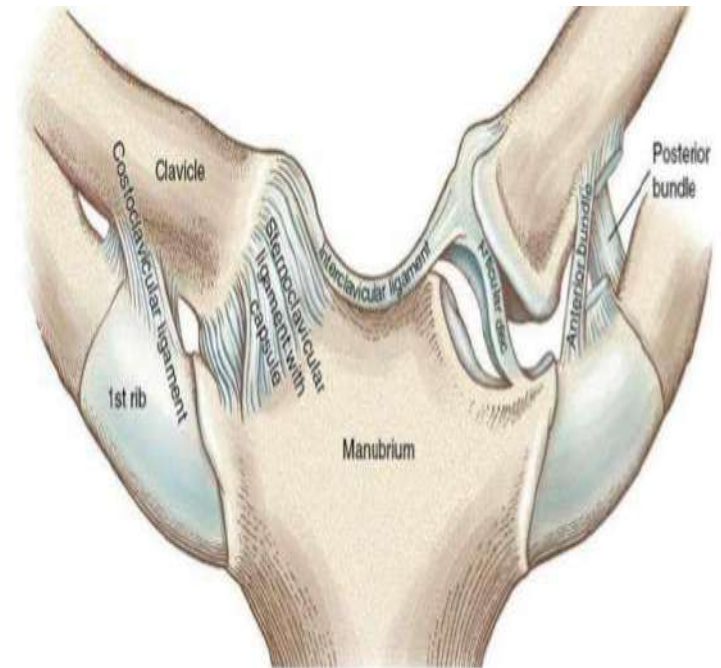
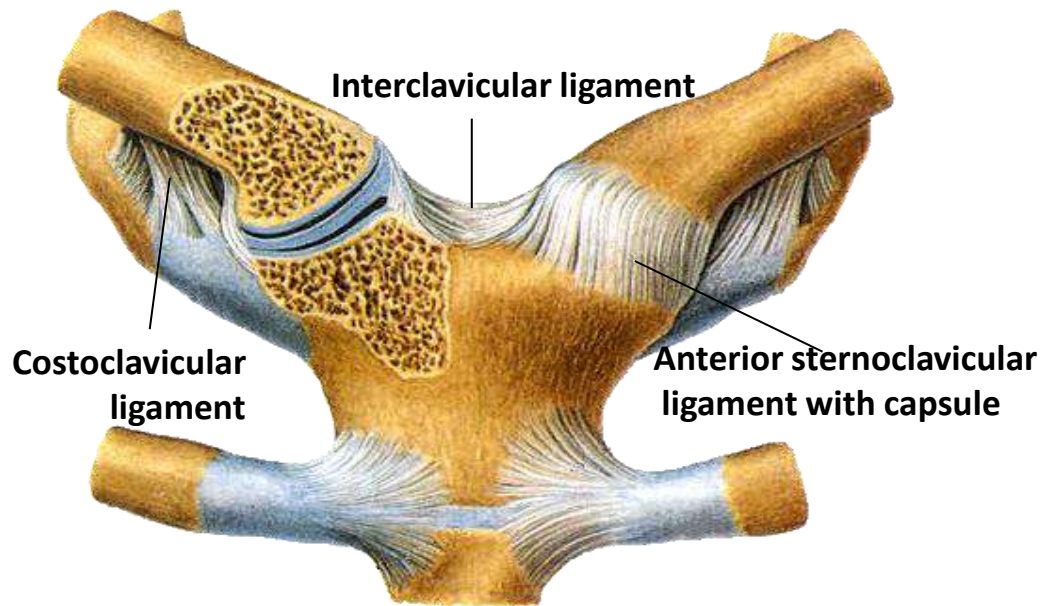
Articular capsule: strong and is reinforced by **anterior and posterior sternoclavicular ligaments** .

Articular disc : is attached to the capsule dividing the joint into 2 cavities .



Ligaments of sternoclavicular joints

1. **Anterior and posterior sternoclavicular ligaments**
2. **Interclavicular ligament** links the ends of the two clavicles to each other and to the superior surface of the manubrium of sternum;
3. **Costoclavicular ligament** is lateral to the joint and links the proximal end of the clavicle to the first rib and related costal cartilage.



Acromioclavicular joint

Type : *Plane synovial joint*

Articular surfaces:

1. Medial border of acromion

2. Lateral end of the clavicle.

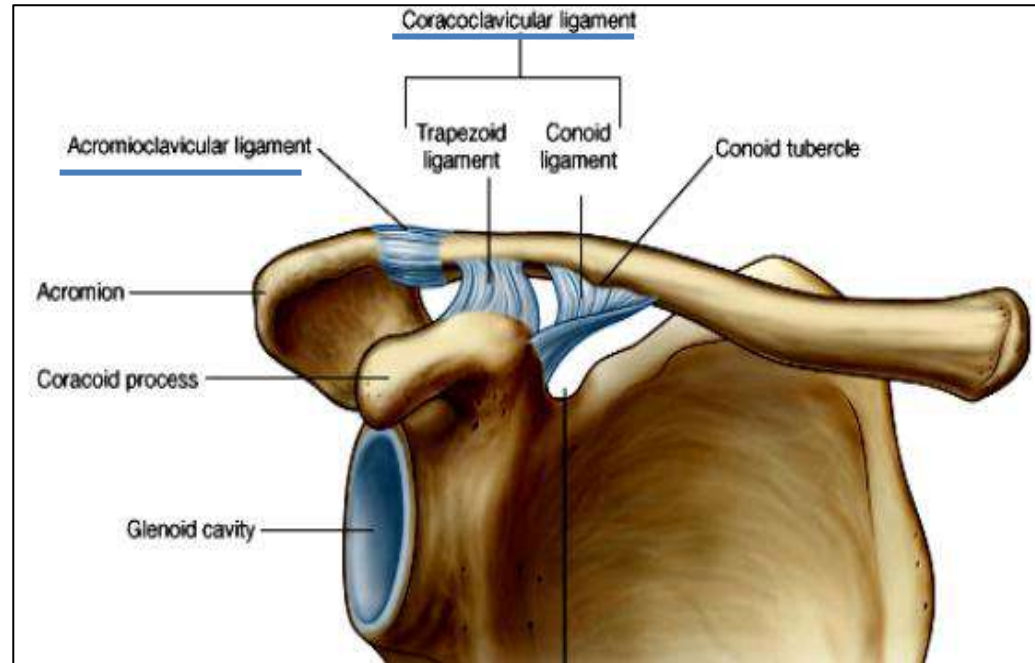
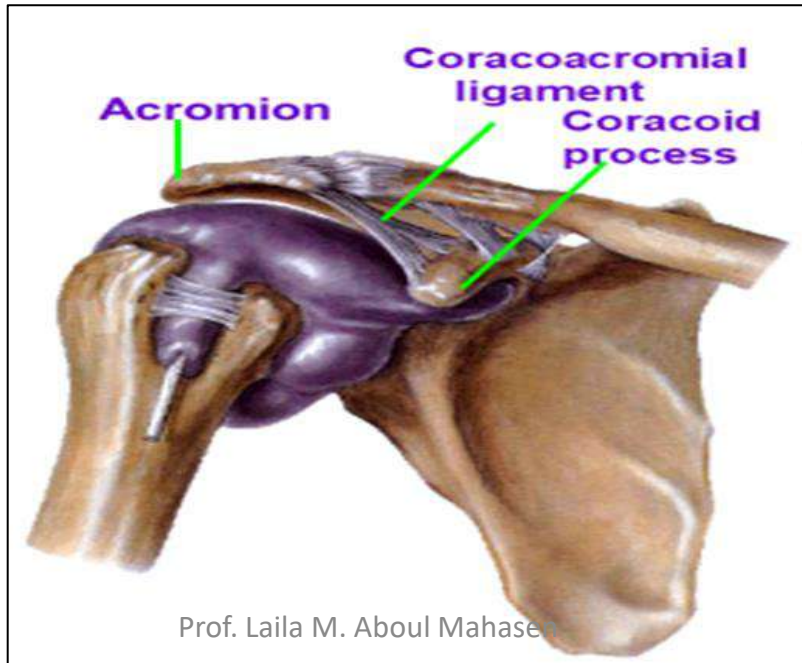
Capsule: attached around the margins of both articular surfaces.

Ligaments : The joint capsule is reinforced by the following ligaments:

1. **Acromioclavicular ligament** superior to the joint

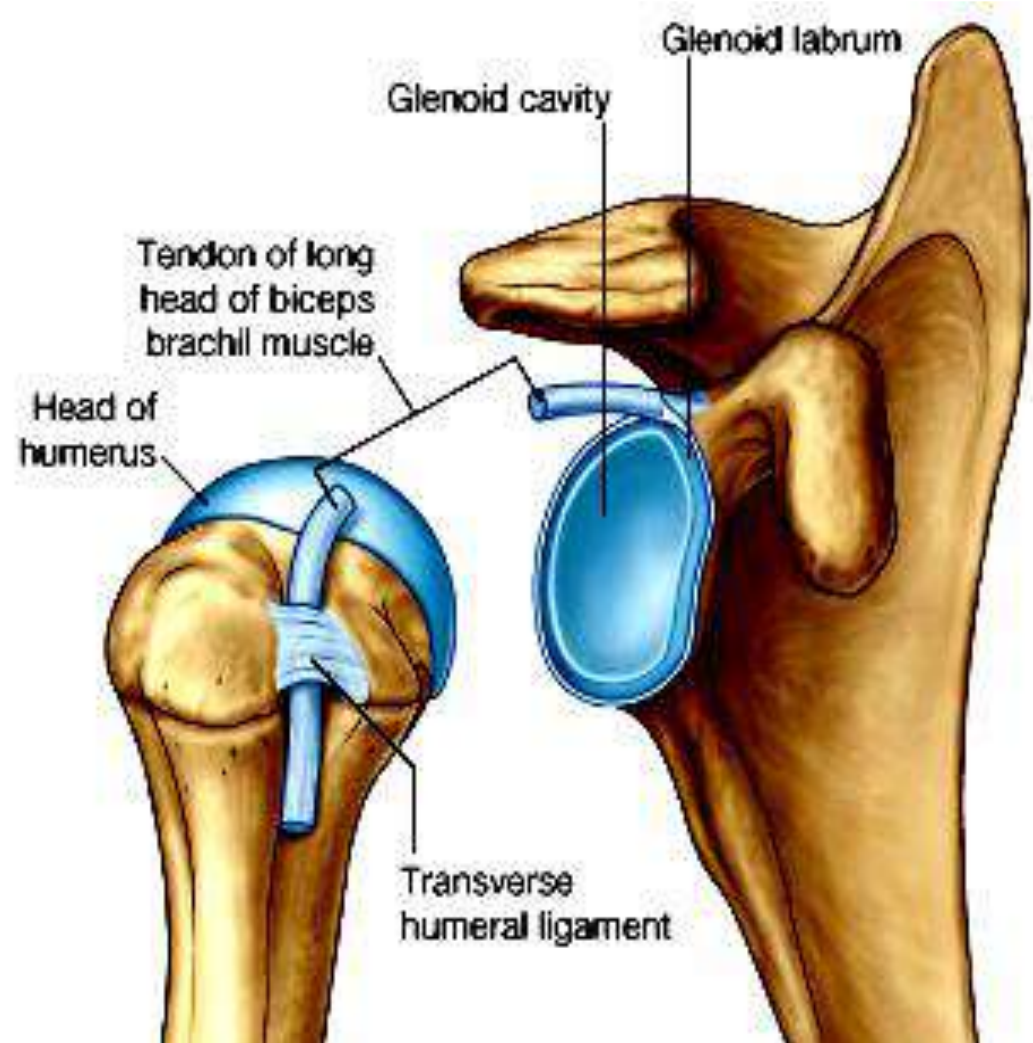
2. **Coracoclavicular ligament** extends from the coracoid process of the scapula to inferior surface of lateral end of the clavicle (it has **trapezoid & conoid parts**).

3. **Coracoacromial ligament** connects the acromion with coracoid process .



Shoulder joint (glenohumeral)

1. Articular surfaces
2. Type of joint
3. Capsule (attachment)
4. Synovial membrane
5. Ligaments
6. Factors strengthen the joint
7. Movements of the joint
8. Nerve supply
9. Dislocation
10. Radiological examination



Shoulder joint (Glenohumeral Joint)

Type:

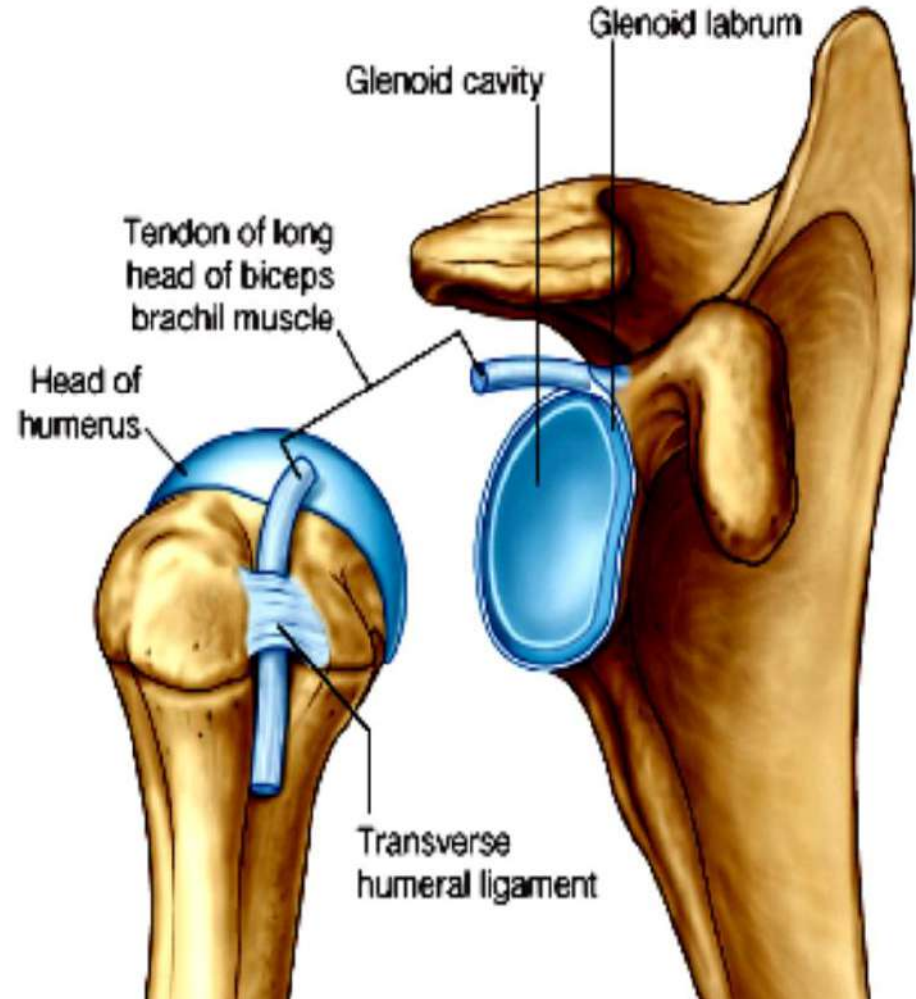
Ball and socket synovial joint

Articular surfaces :

- 1. The glenoid cavity of the scapula**
- 2. The head of the humerus.**

Both articular surfaces are covered with **hyaline cartilage**.

The shallow glenoid cavity is **deepened** by a fibrocartilaginous **glenoid labrum**



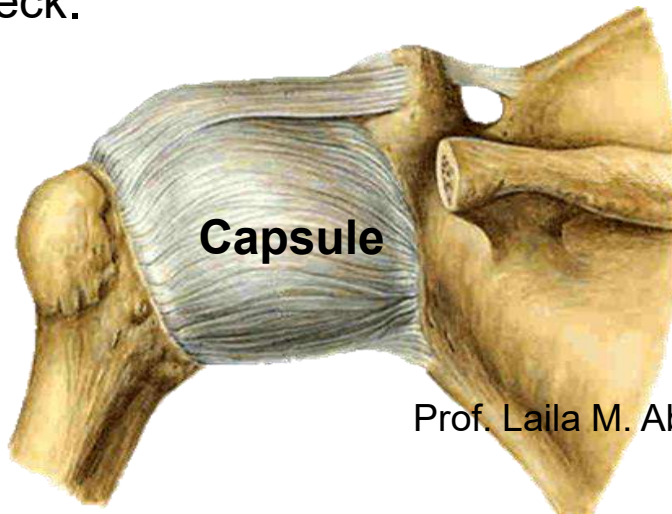
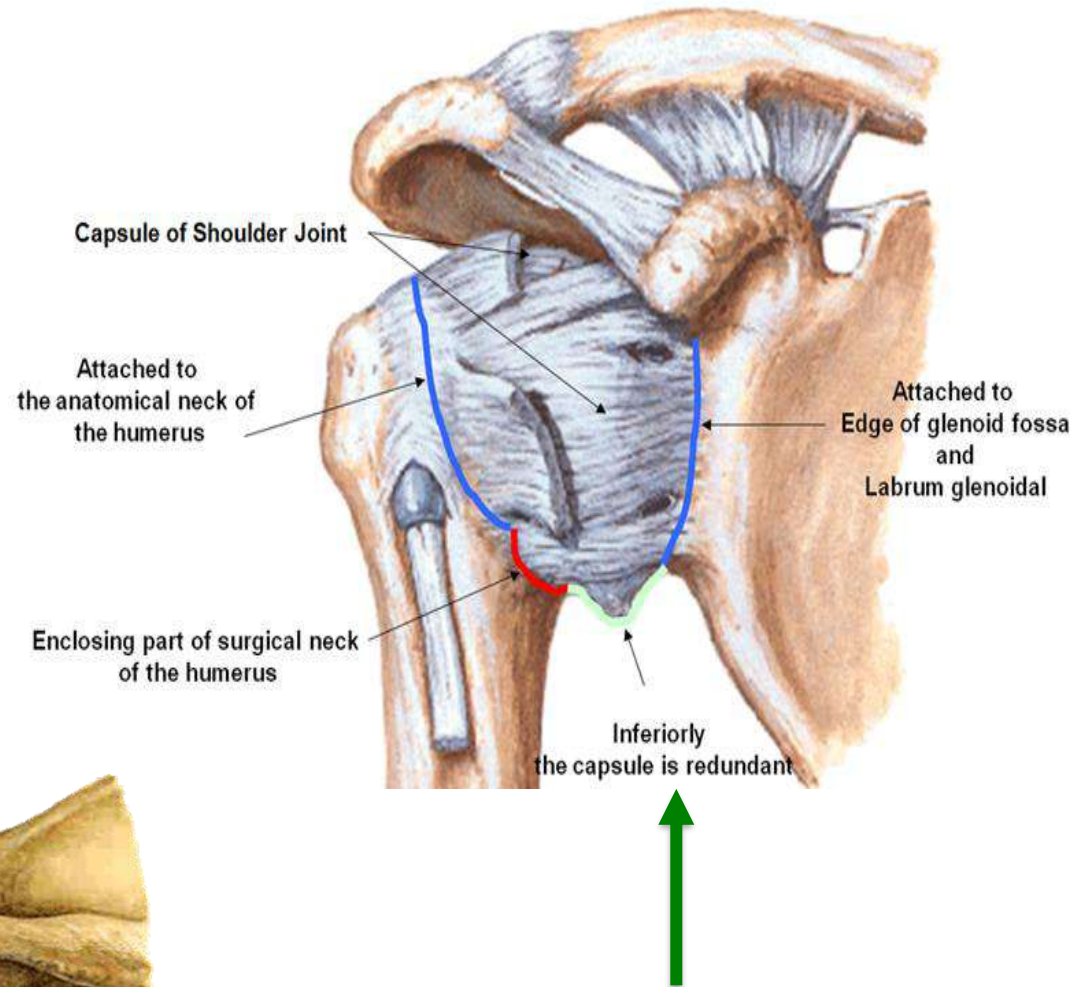
Capsule of shoulder joint

It is thin & lax allowing a wide range of movement .

Attachment :

Superiorly : to the margin of the glenoid cavity outside the glenoid labrum and **above the supraglenoid tubercle** (so the biceps tendon is intracapsular)

Inferiorly : to anatomical neck of t humerus except medially it is attached 1\2 an inch below the anatomical neck.



Factors strengthen the joint

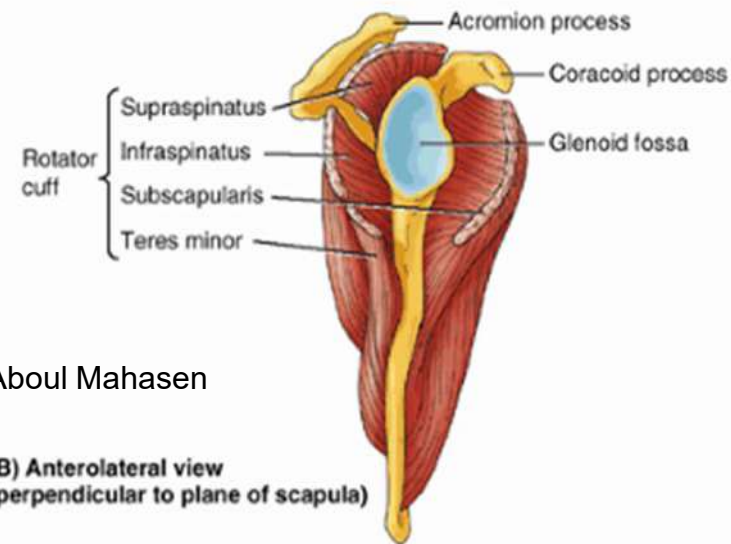
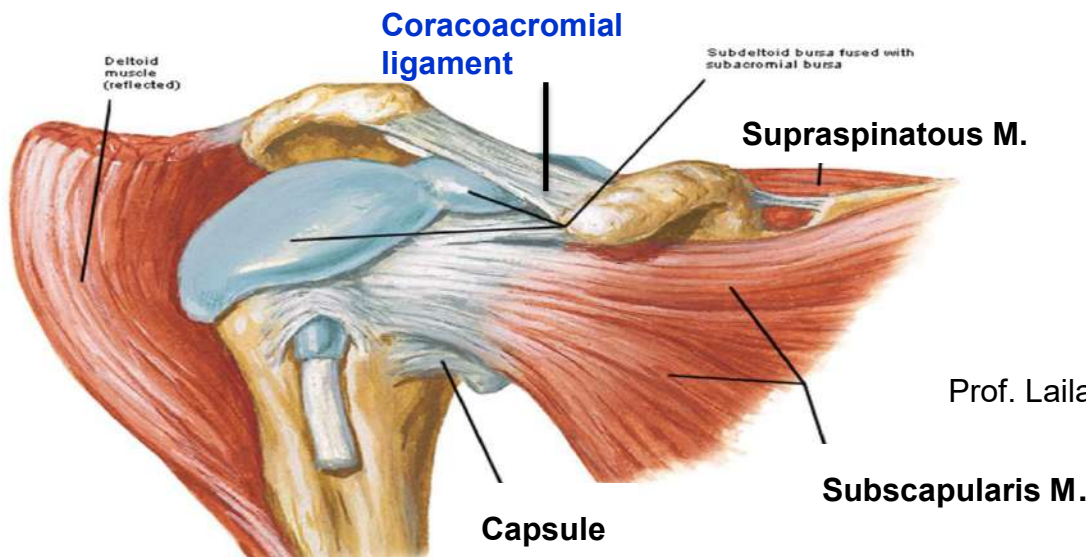
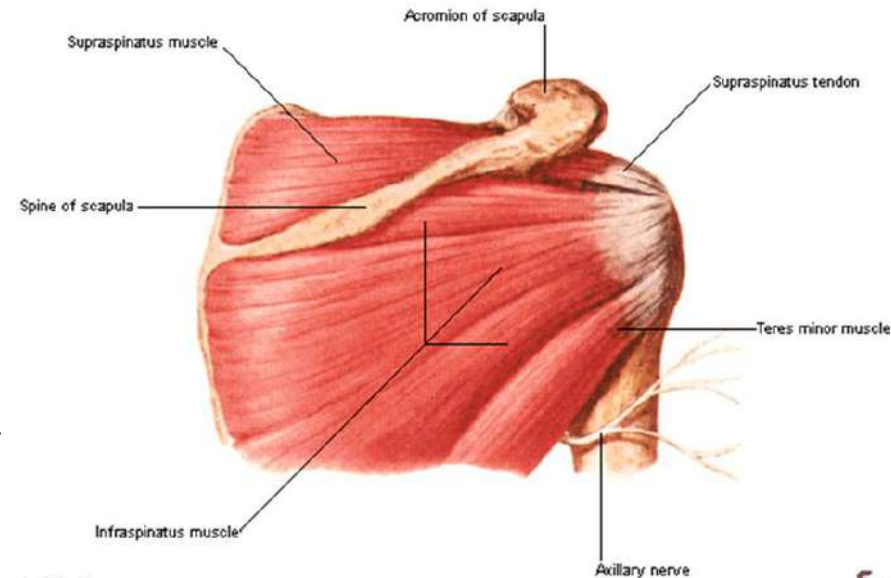
The capsule is reinforced by :

1- The rotator cuff which is formed by :

- a. Subscapularis (anteriorly) ,
- b. Supraspinatus (superiorly)
- c. Infraspinatus and Teres minor (Posteriorly).

2- The coracoacromial arch is composed of (coracoacromial ligament , coracoid process & acromion) forming a **2ry socket** for the head of the humerus .

Muscles of Rotator Cuff
Posterior View



Prof. Laila M. Aboul Mahasen

(B) Anterolateral view
(perpendicular to plane of scapula)

Ligaments of Shoulder joint

I. Glenohumeral ligaments :

They are **3 ligaments** (superior , middle & inferior) , they **strengthen the anterior part of the capsule**.

II. . Transverse humeral ligament

It extends between the greater and lesser tubercles.

It holds the **tendon of the long head of the biceps** in the intertubercular groove.

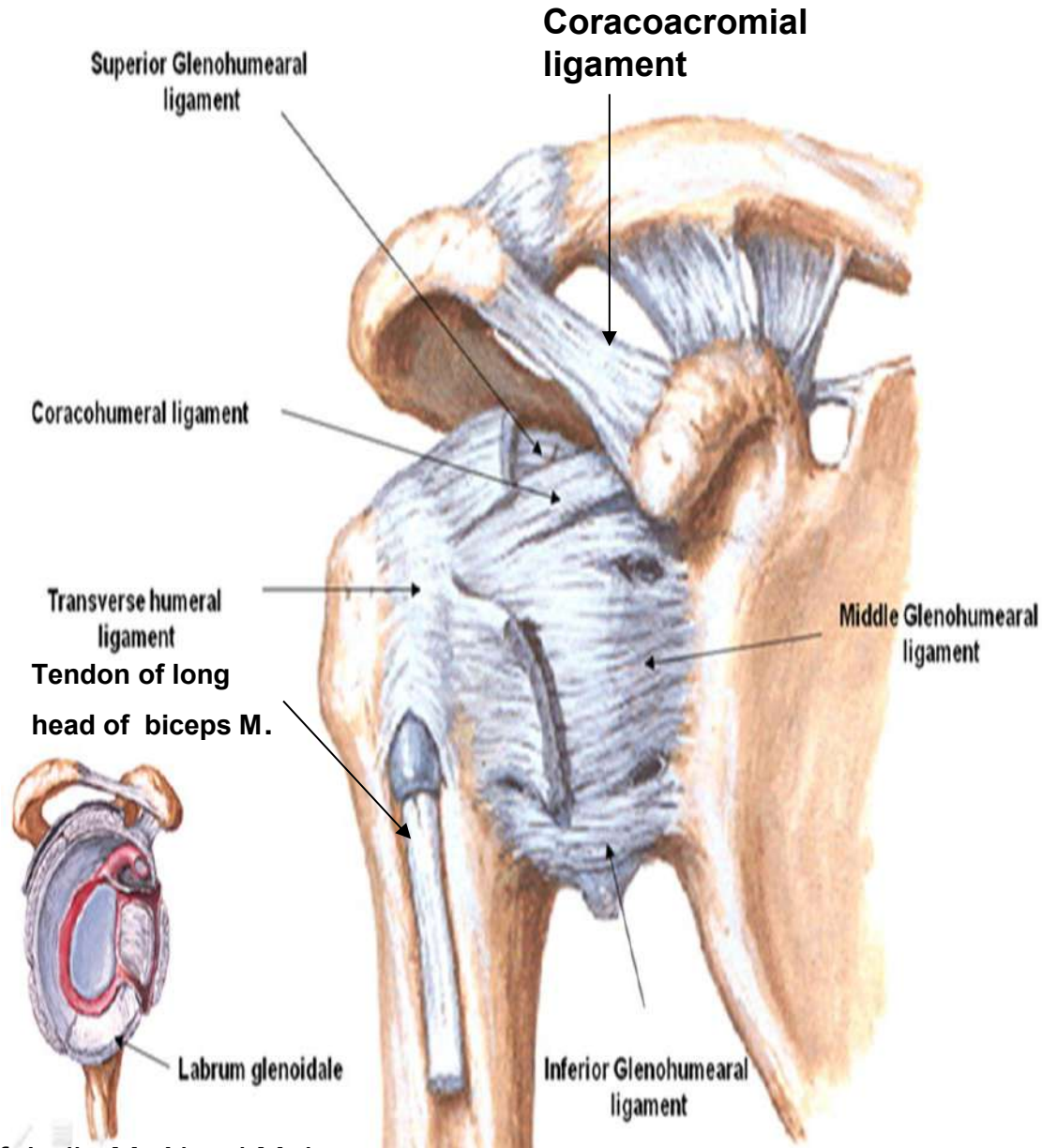
III. Coracohumeral ligament :

It strengthens the upper part of the capsule).

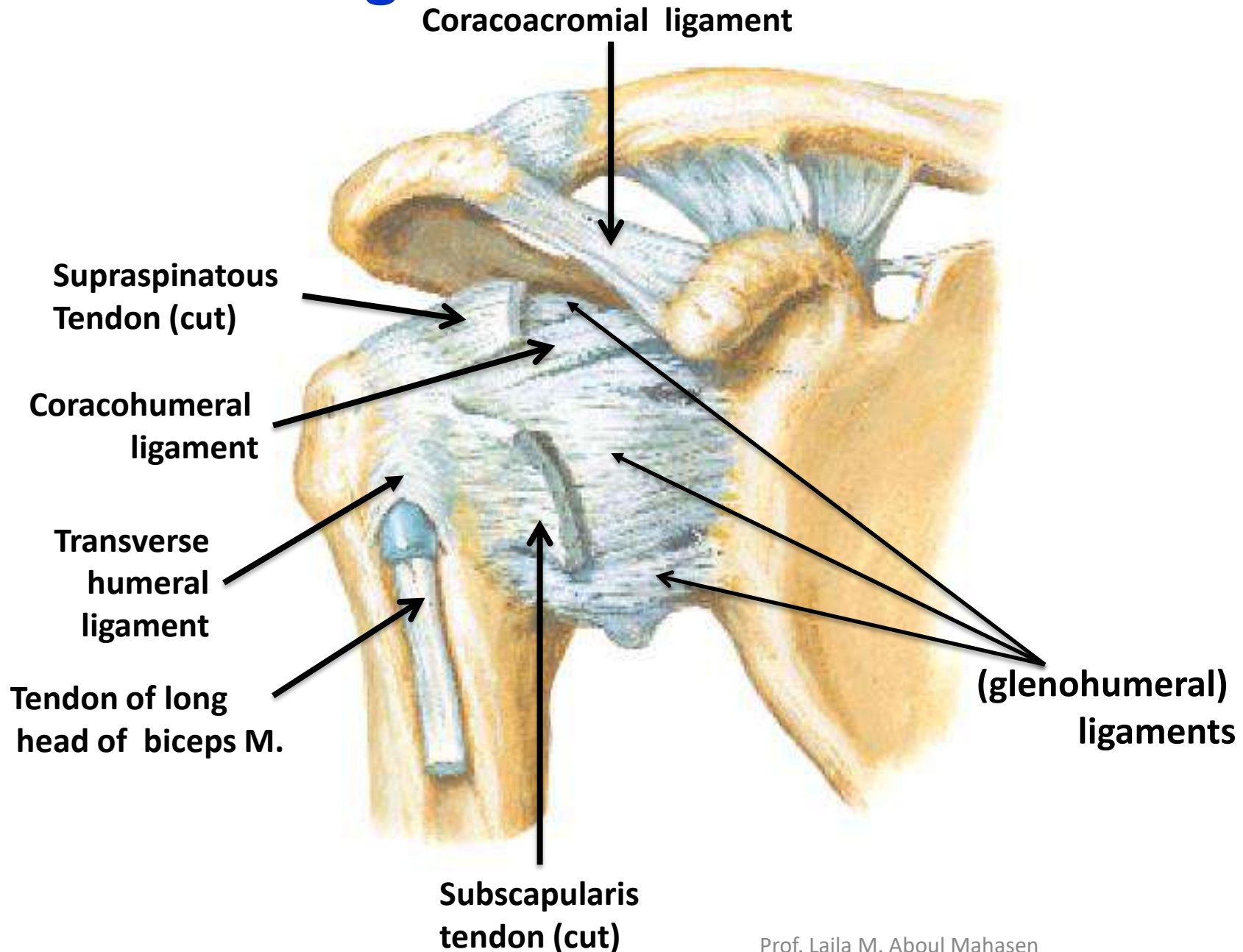
It extends from the coracoid process to the greater tubercle of the humerus

IV. Coracoacromial ligament :

It protects the upper part of the joint . It extends from the coracoid process to the acromion.



Ligaments of Shoulder Joint



Muscles producing movements of shoulder joint

Adduction : pectoralis major, latissimus dorsi & teres major

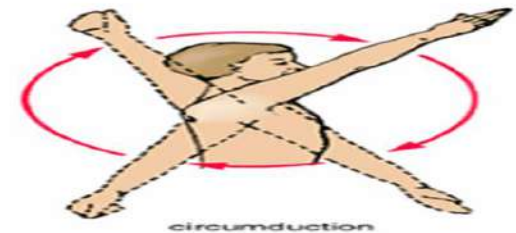
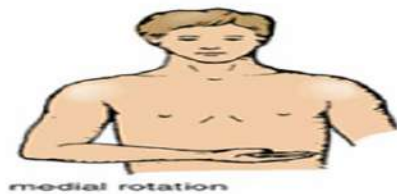
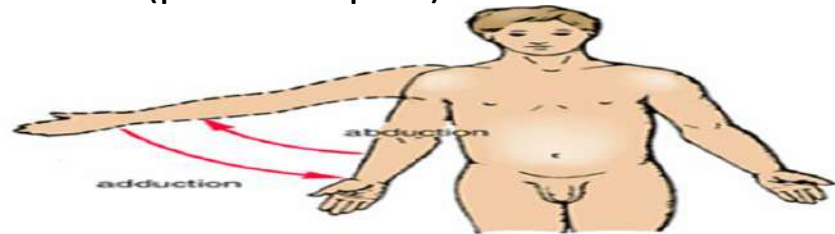
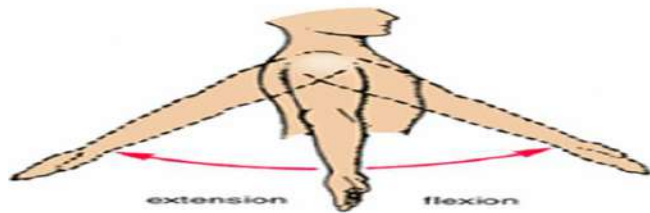
Abduction : middle fibres of deltoid (15- 90) & supraspinatus(0-15) , above 90 by serratus anterior .

Flexion : pectoralis major (clavicular part) , deltoid (anterior part) & Coracobrachialis , biceps .

Extension : latissimus dorsi, deltoid (posterior part) .

Medial rotation: subscapularis, pectoralis major, deltoid (anterior part), latissimus dorsi & teres major .

Lateral rotation : Infraspinatus, teres minor, deltoid (posterior part)

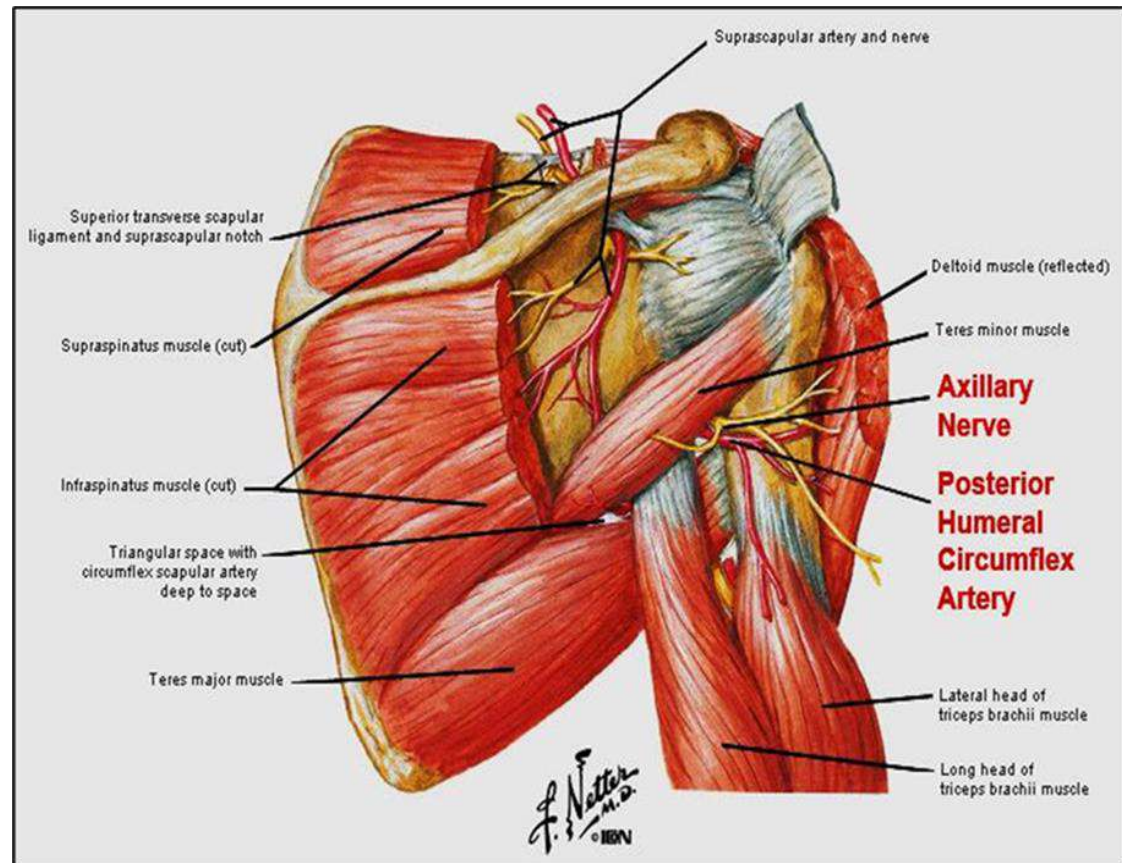


Nerve supply of shoulder joint

Hilton's Law states that the same nerves that supply the muscles moving a joint and the skin over those muscles also supply the interior of the joint itself .

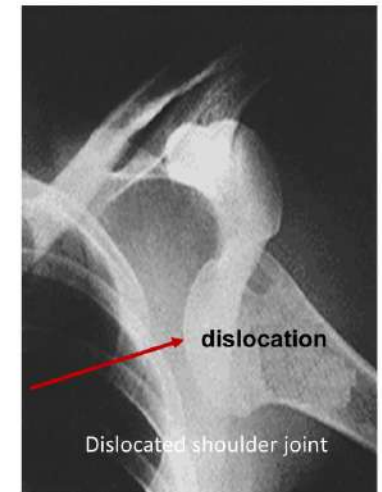
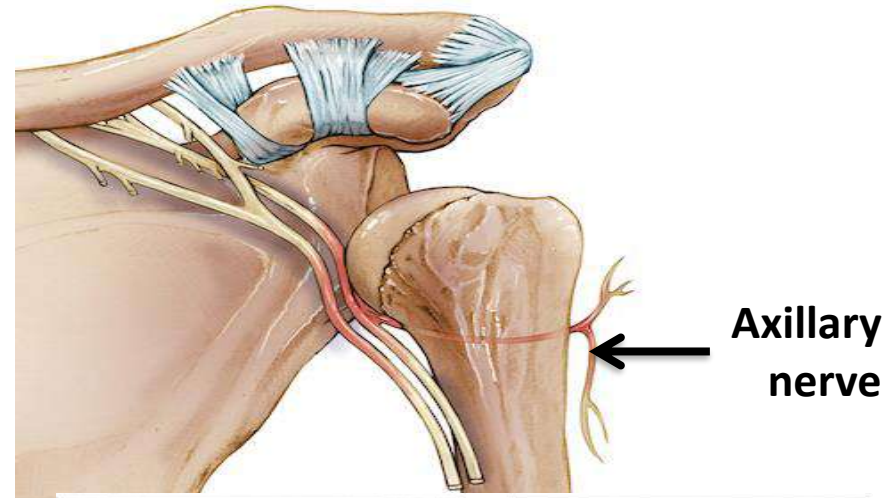
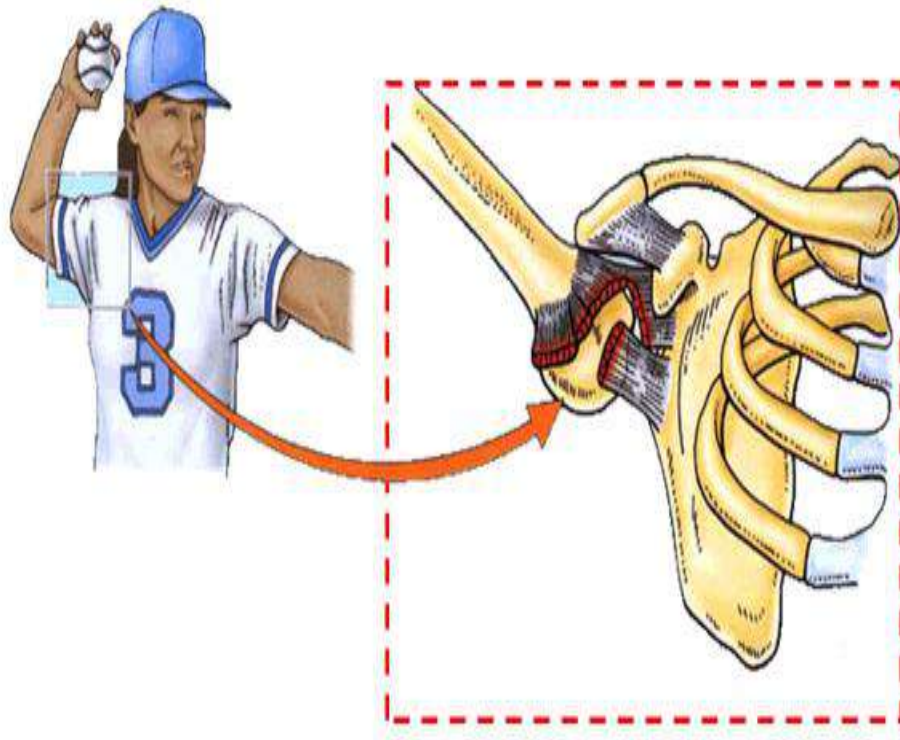
Nerve supply of the shoulder joint :

1. Suprascapular nerve
2. Axillary nerve
3. Musculocutaneous nerve
4. Medial & lateral pectoral nerves



Dislocation of the shoulder joint

It may be subject to **inferior or anterior dislocation**, which stretches the fibrous capsule, avulses the glenoid labrum, and may **injure the axillary nerve** in quadrangular space .



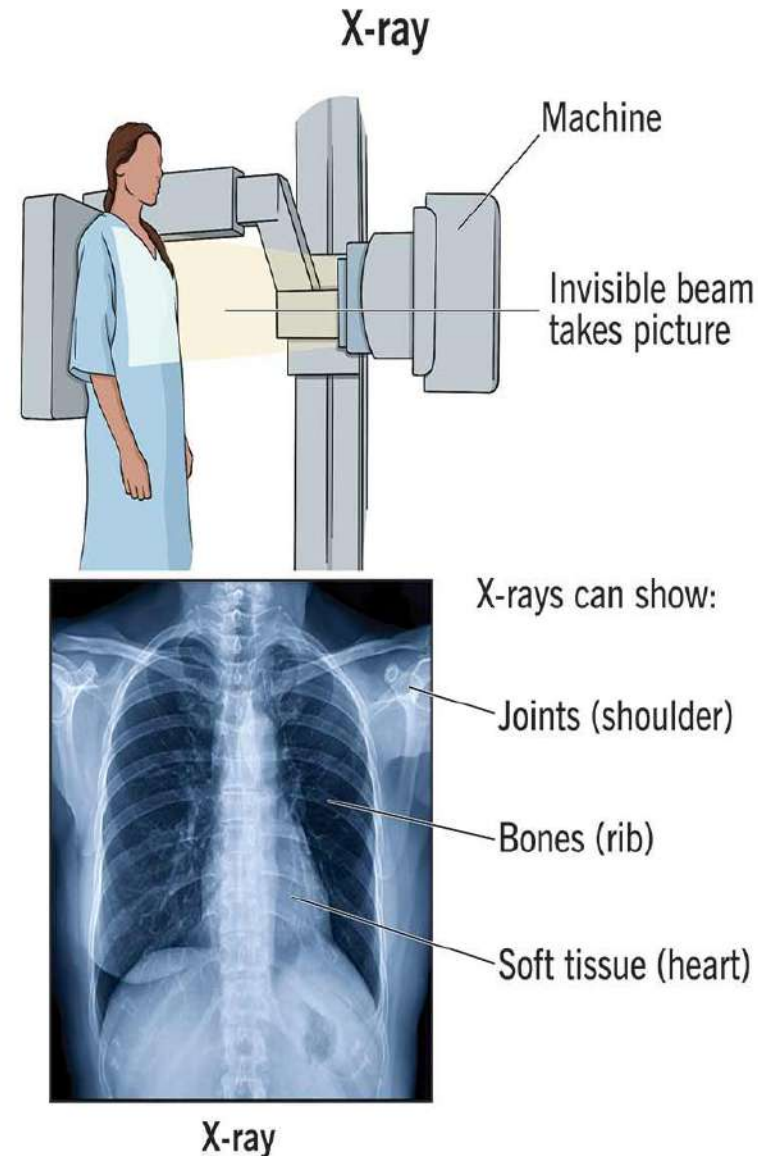
Radiological examination of Upper limb joints

What is Radiology?

❑ It is a medical specialty that uses imaging to diagnose and treat diseases seen within the body.

Common imaging techniques of skeletal system

1. X-ray radiography : for bone fractures , degeneration or joint dislocations
2. computed Tomography (CT): for bone pathology . It offers detailed images, But **NOT** sensitive as **MRI** for certain bone conditions. .
3. Magnetic Resonance Imaging (MRI): for soft tissues pathology like tumors & metastasis . **Most sensitive and specific technique**



X-ray Radiography

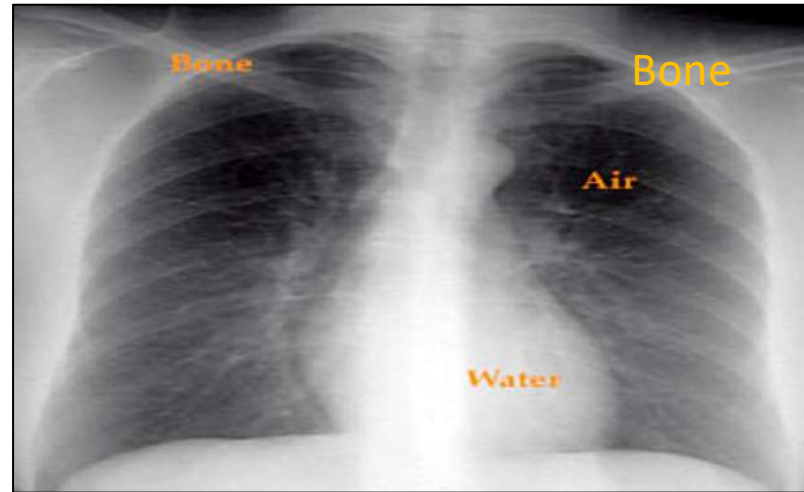
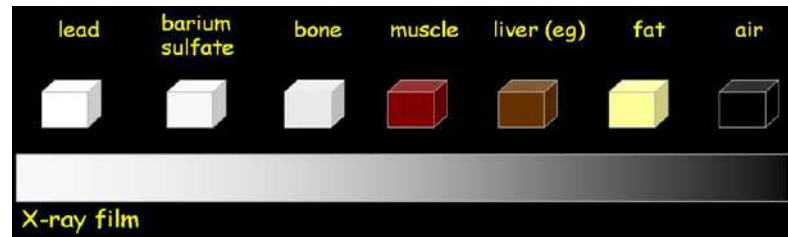
X- Ray Densities :

The big two densities are:

- (1) **WHITE** : Bone
- (2) **BLACK** : Air

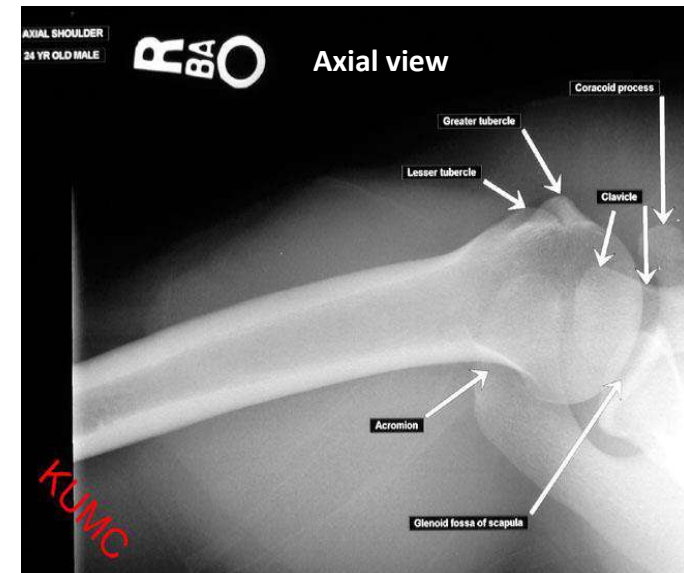
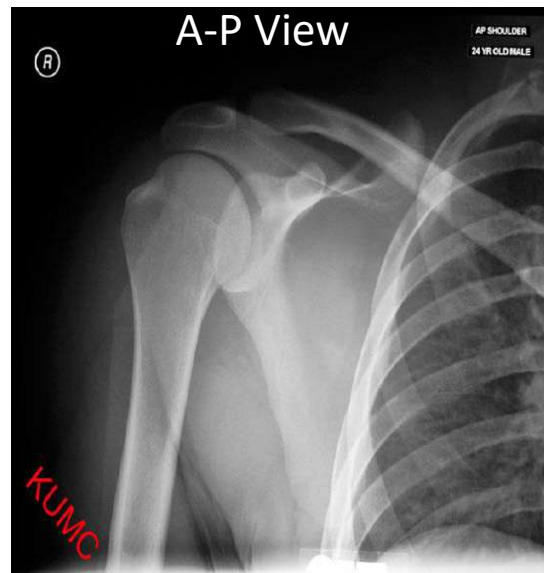
The others are:

- (3) **DARK GREY**- Fat
- (4) **GREY**- Soft tissue/water



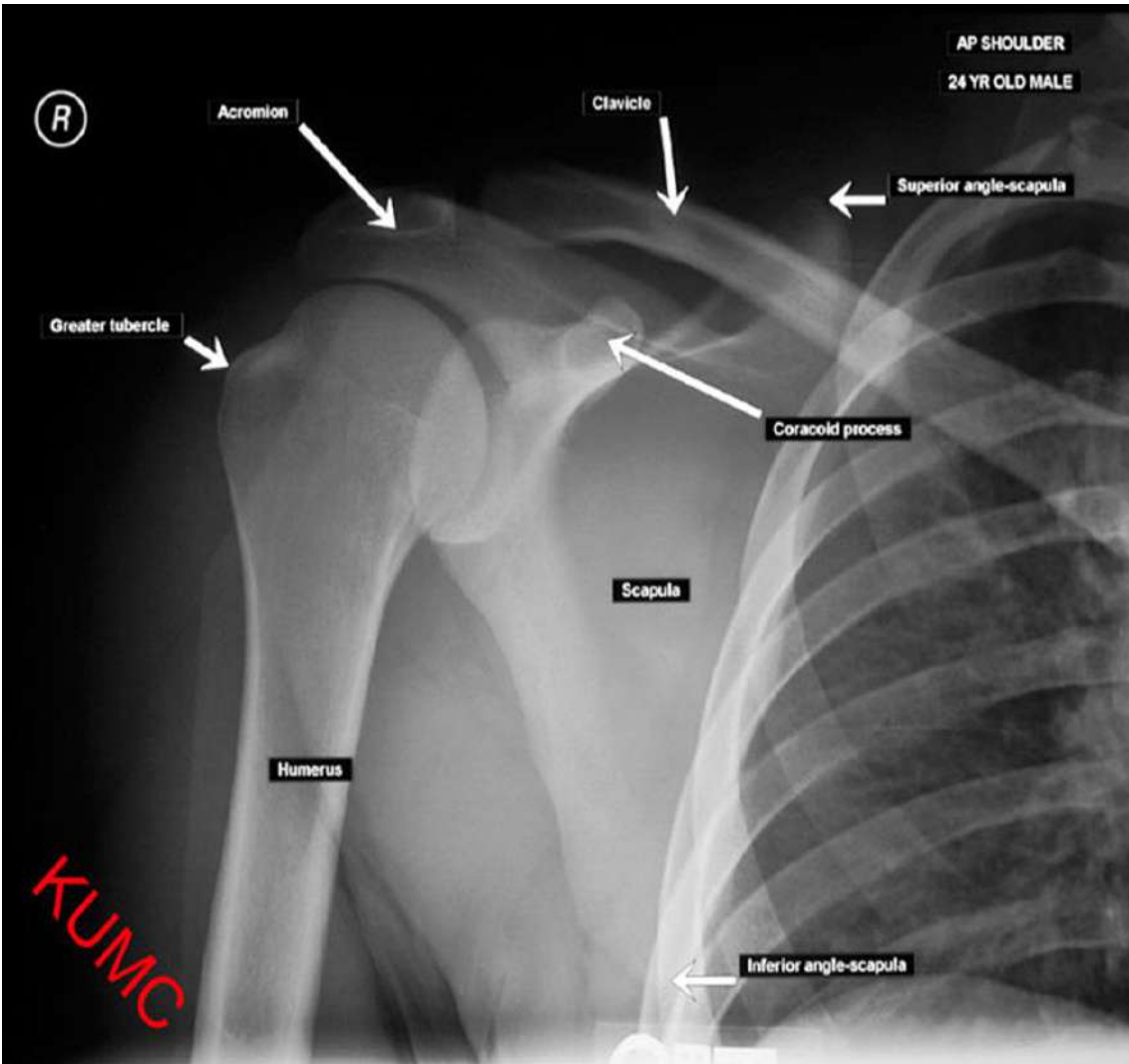
X-Ray Views :

- AP : Anteroposterior
- PA : Postero-anterior view
- Lateral view
- Axial view

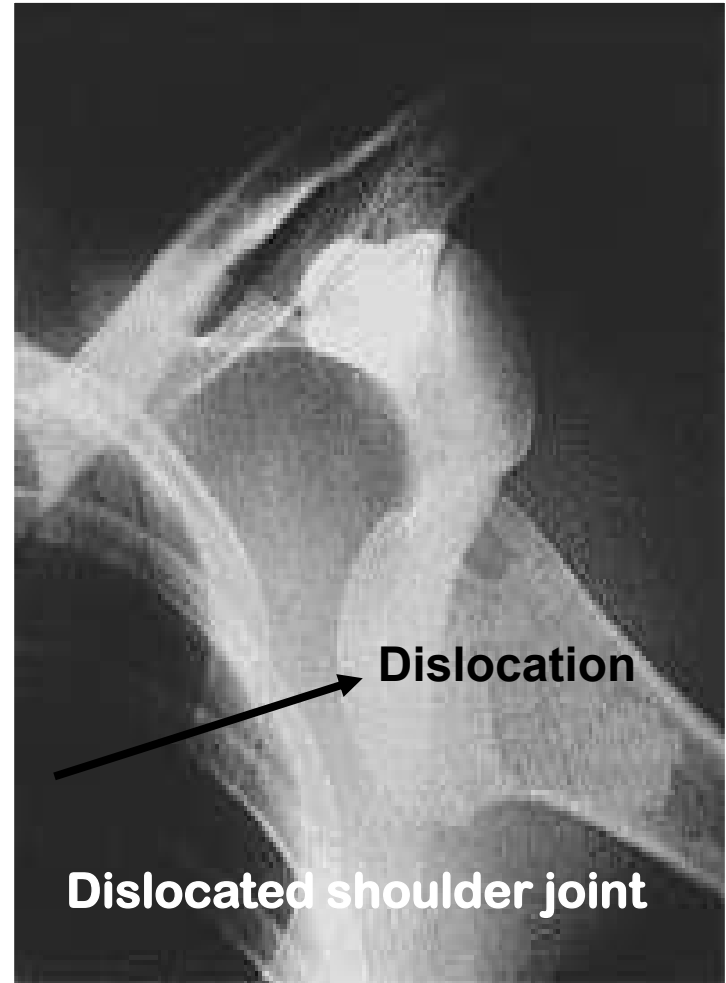


A-P View

Shoulder Joint



Inferior Dislocation of shoulder joint

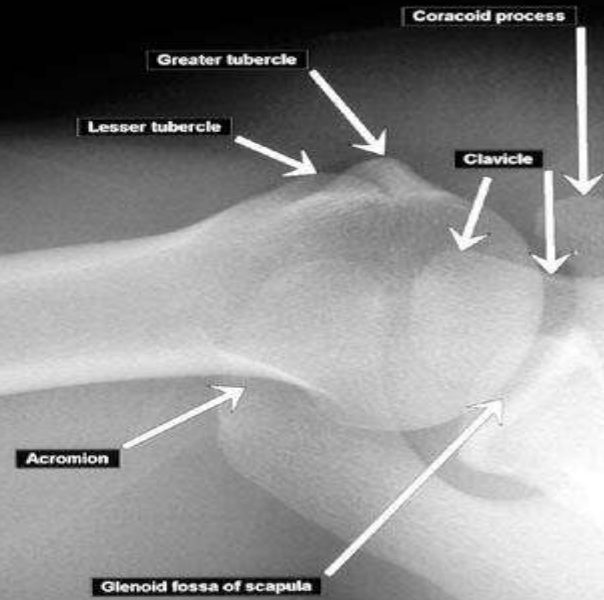


AXIAL SHOULDER
24 YR OLD MALE

RBAO

Axial V.

KUMC



RBAO

24 YR OLD MALE

KUMC

AP –View of abducted shoulder joint(15 degrees)

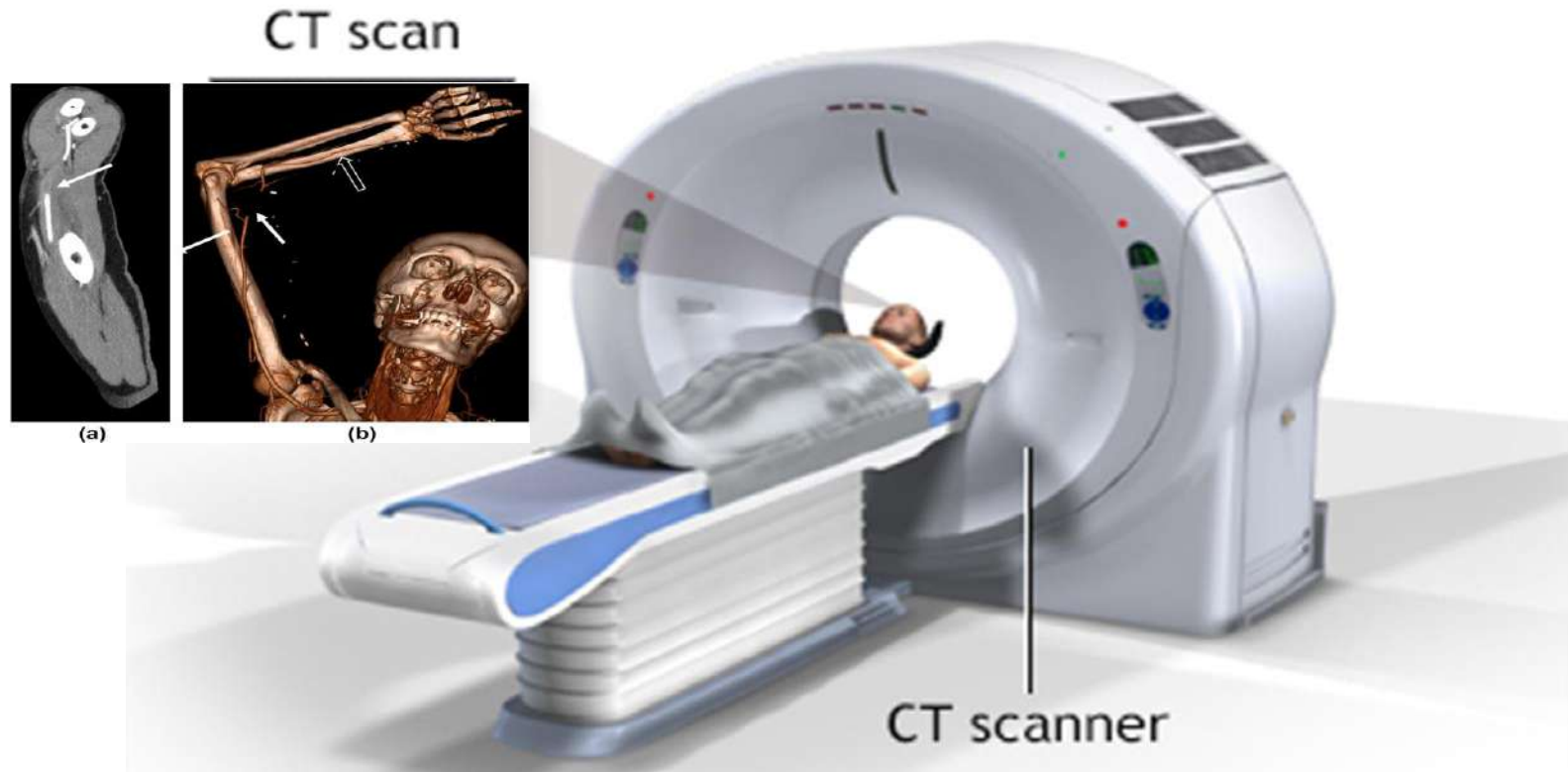


Computed Tomography (CT)

Technique :

CT relies on the same physics as conventional x-rays, in that structures are distinguished from one another by their ability to absorb energy from x-rays.

The information is fed to a computer that processes the information, which is then displayed as a reconstructed picture on a computer . Essentially, the observer sees an **image of a thin slices through the bones (horizontal –sagittal –vertical) ,** which may then be photographed for later examination .



Computed Tomography (CT)

- Widely available
- **Excellent bone assessment** But **NOT** give details for muscles & joints as **MRI**
- Patient friendly & Moderate expense



(a)

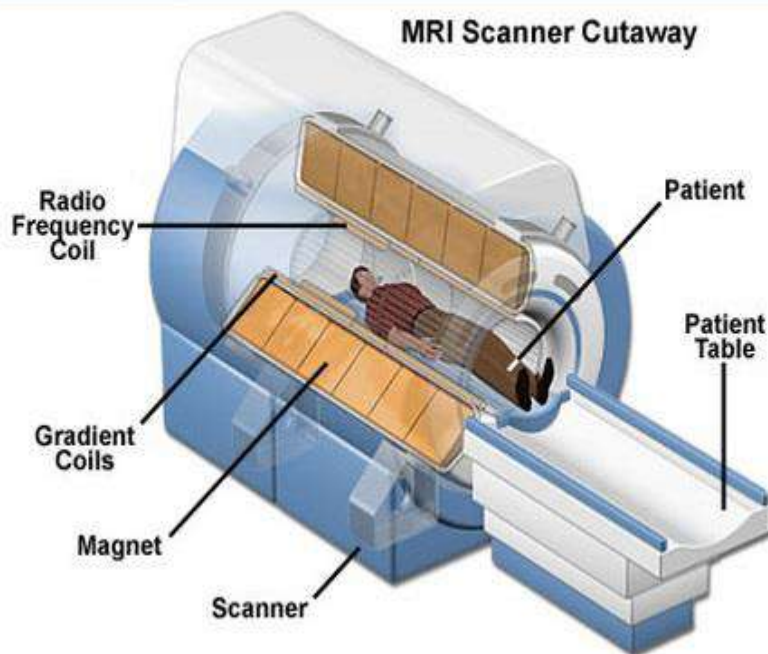


(b)

Magnetic Resonance Imaging (MRI)

- Widely available, but non-standardized imaging techniques
- **Diagnostic for soft tissue pathology as tumors & metastasis**
- **Excellent for bone pathology**
- NOT patient friendly & Large expense

What is MRI machine? How Does It Work?



Coraco-acromial ligament

supraspinatus

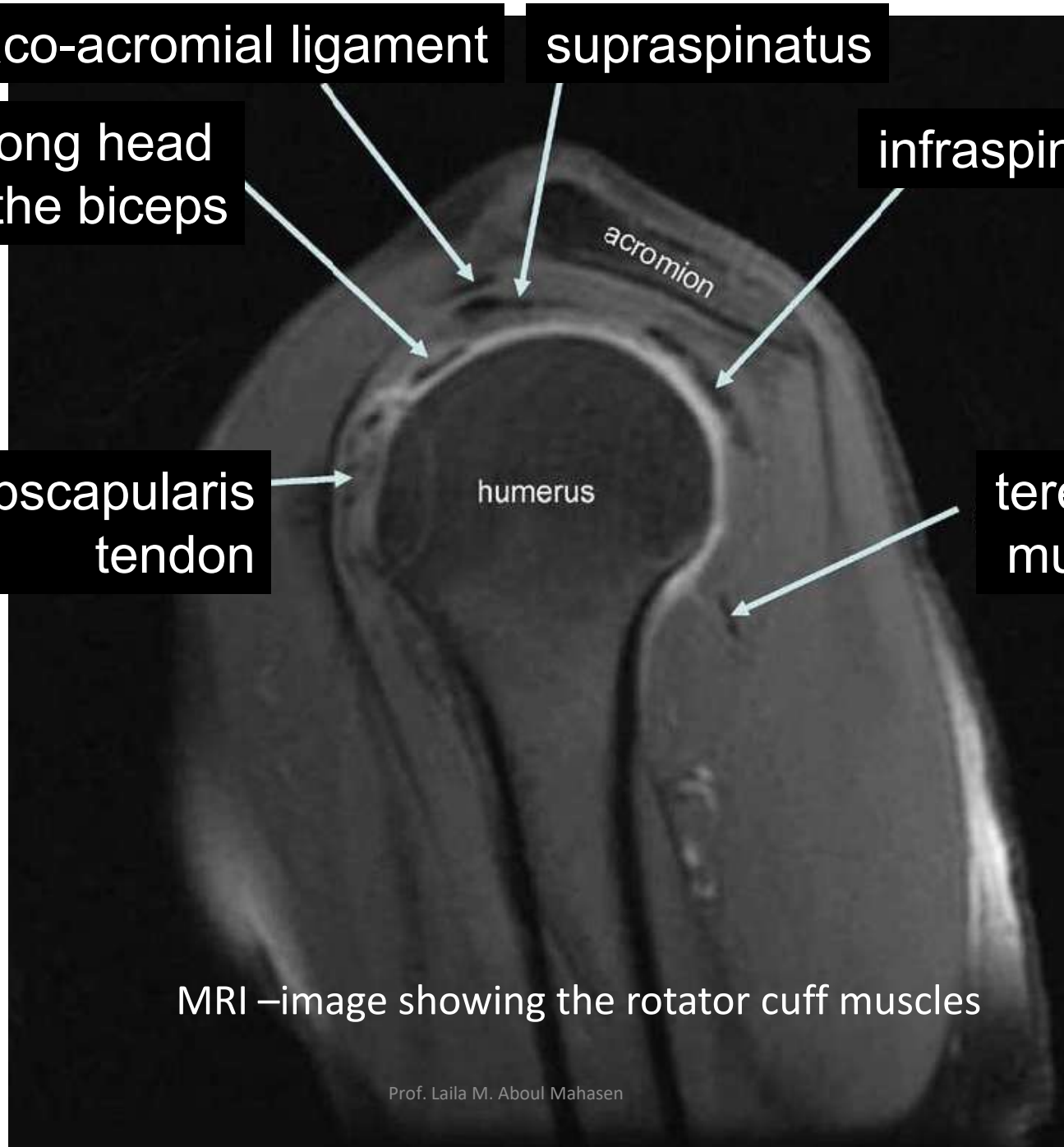
long head
of the biceps

infraspinatus

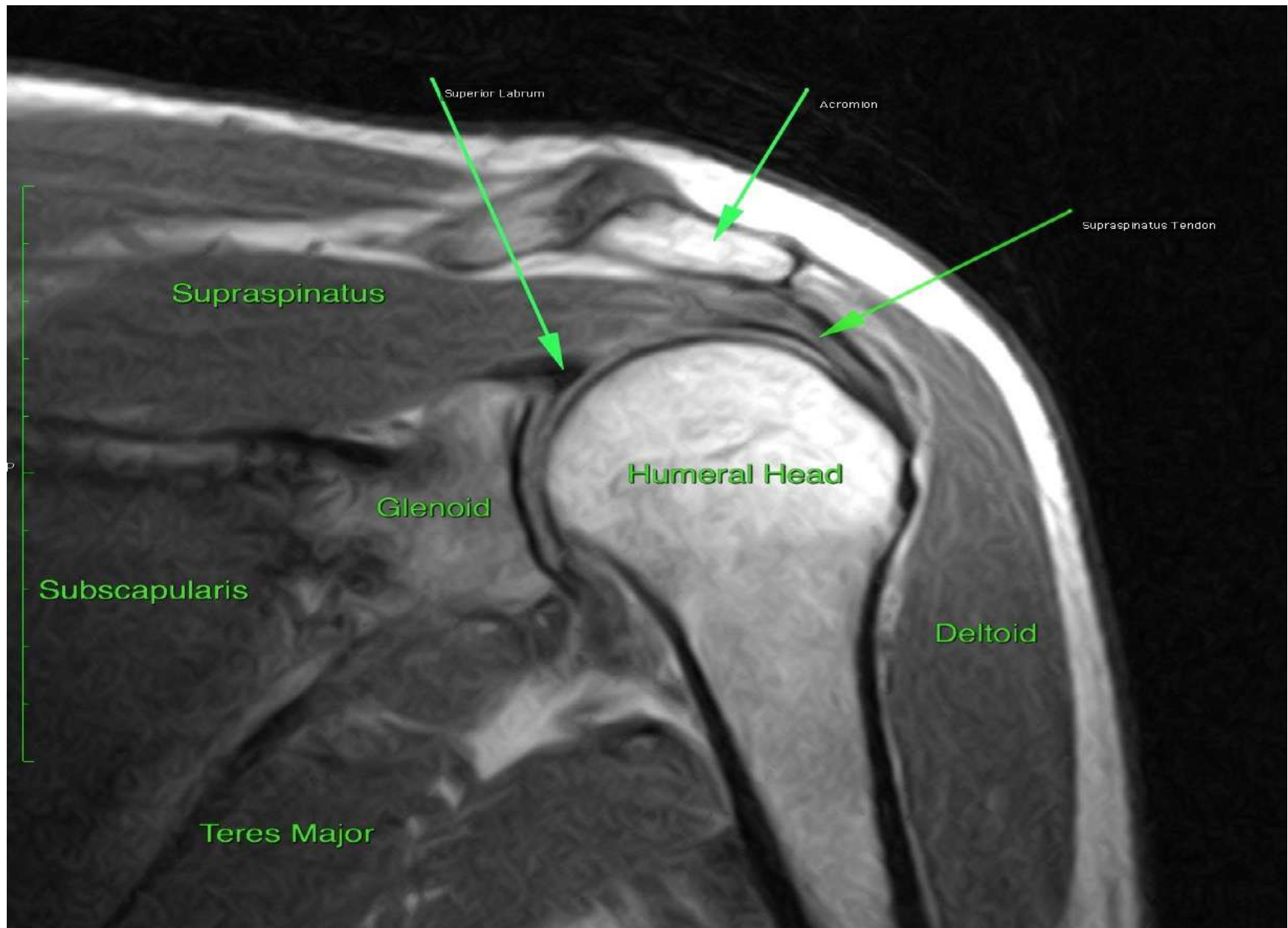
subscapularis
tendon

teres minor
muscle

MRI –image showing the rotator cuff muscles



MRI –image showing the shoulder region (Joint & muscles)



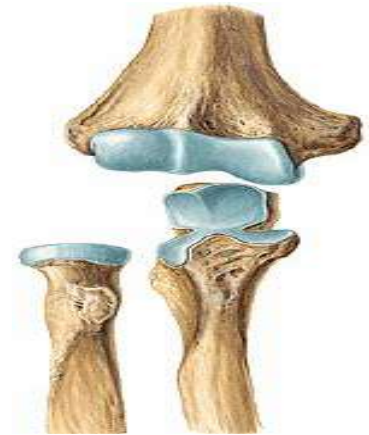
ELBOW JOINT

Type : Synovial hinge joint

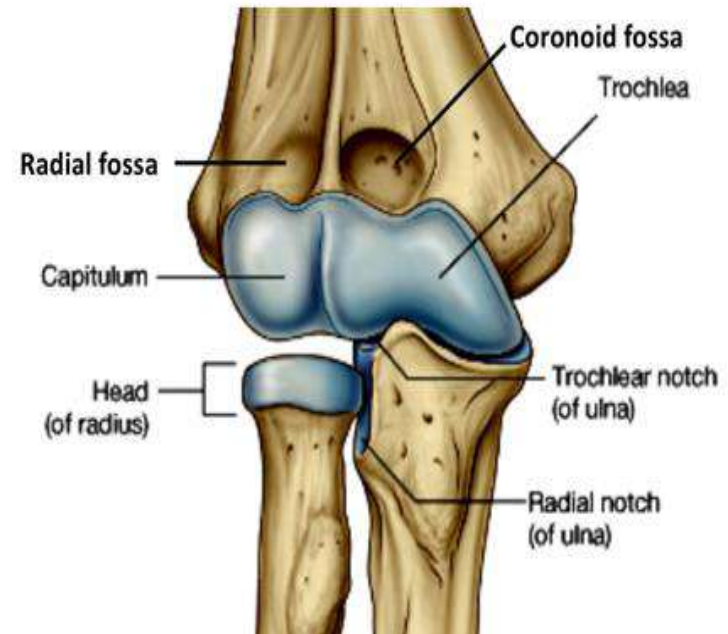
Articular surfaces:

Proximally: trochlea & capitulum of humerus

Distally: trochlear notch of ulna & head of the radius

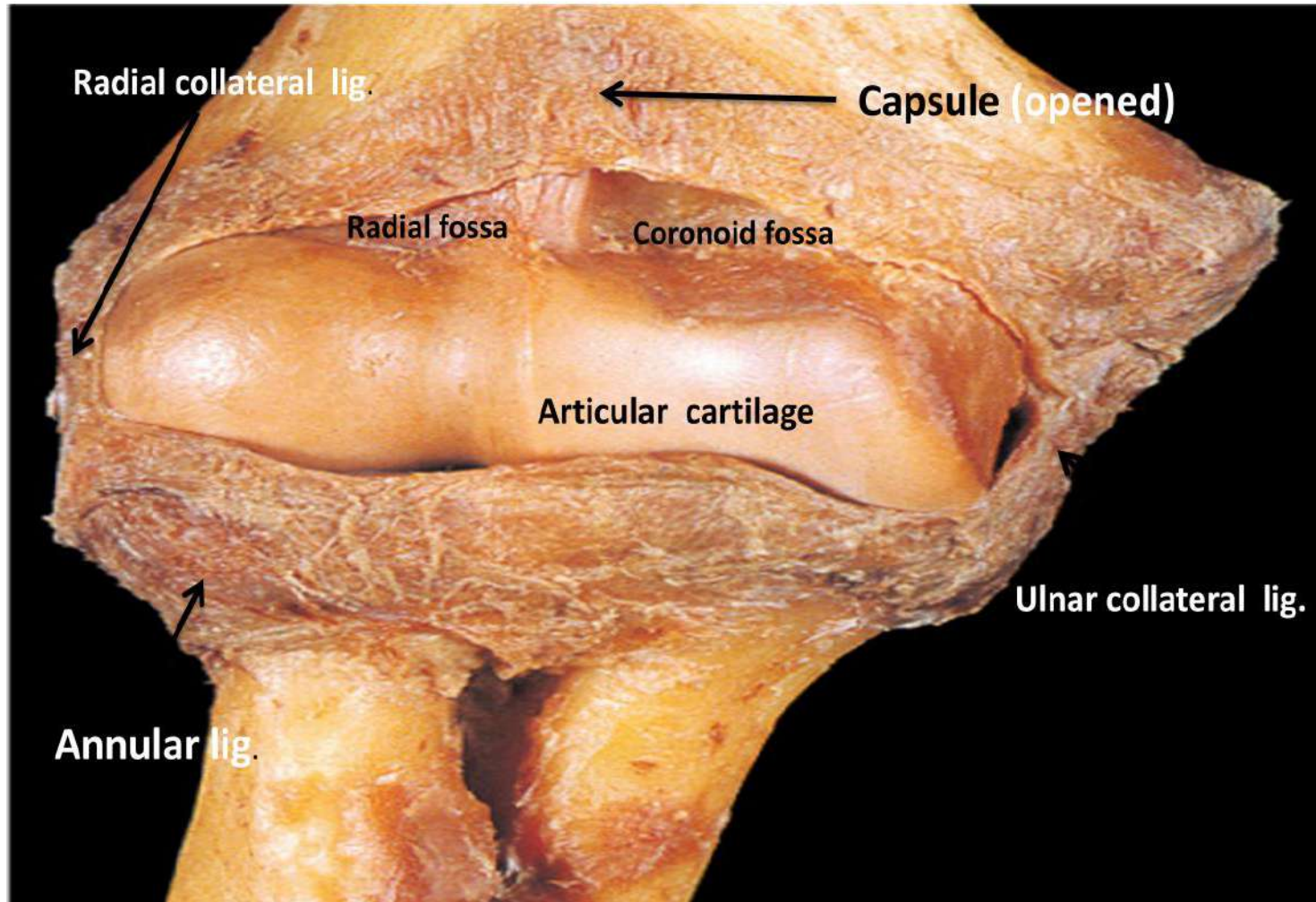


The articular surfaces are covered with **hyaline cartilage**



Anterior view

Anterior view of **elbow joint** to show the capsule , ligaments & the articular cartilage of trochlea through the opened capsule



Fibrous Capsule

The fibrous capsule completely encloses the joint.

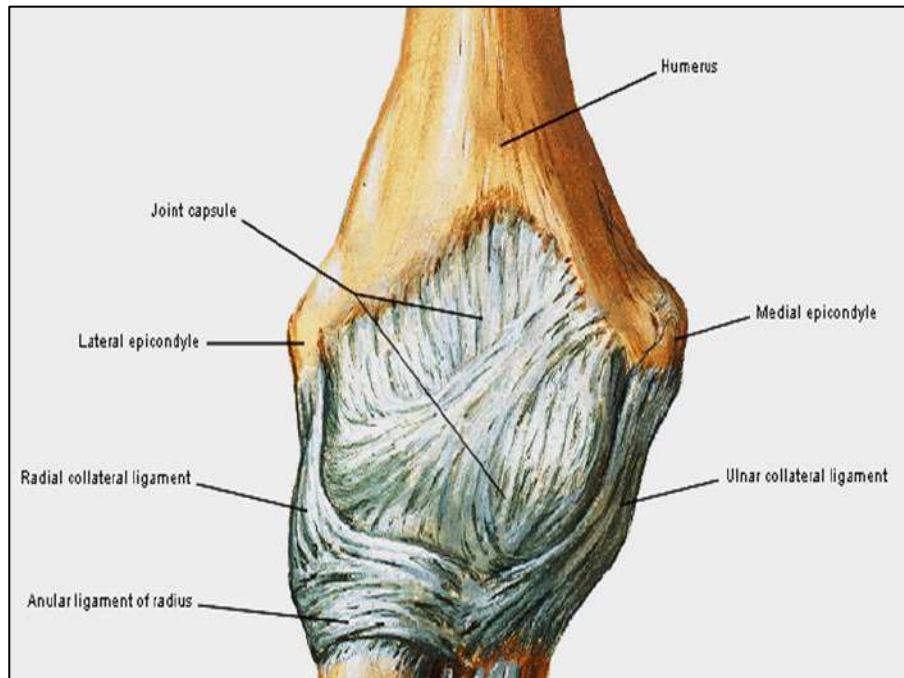
Its anterior and posterior parts are thin and weak, but two thick **collateral ligaments** strengthen its sides.

The fibrous capsule is attached to :

Proximally & anteriorly to the front of the humerus just above the coronoid and radial fossae

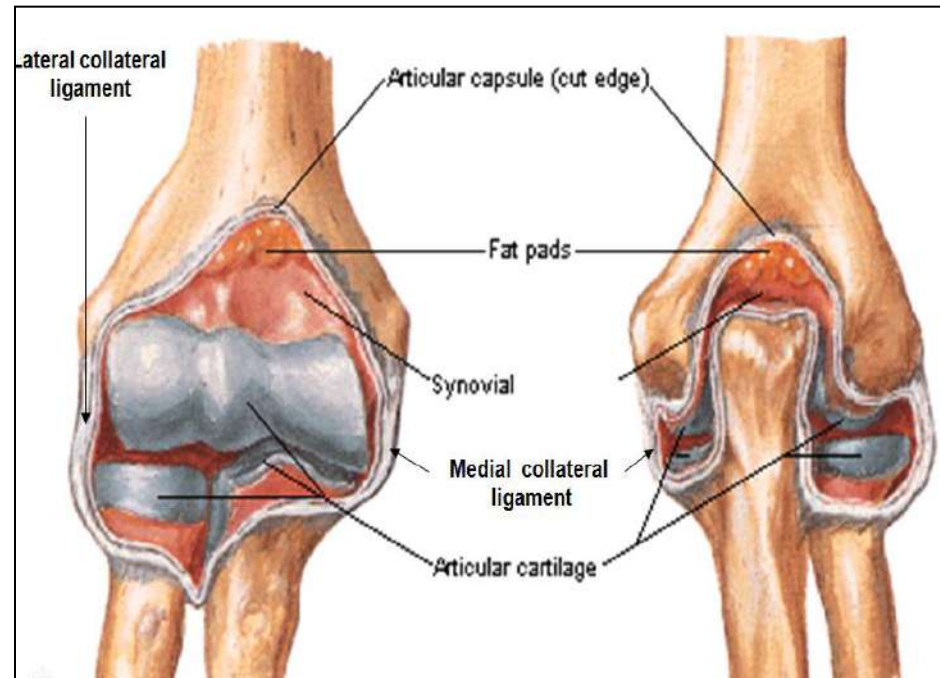
Proximally & posteriorly attached to the back of humerus just above the olecranon fossa.

Distally :the fibrous capsule is attached to the margins of the trochlear notch, the anterior border of the coronoid process and the annular ligament .



The capsule & collateral ligaments

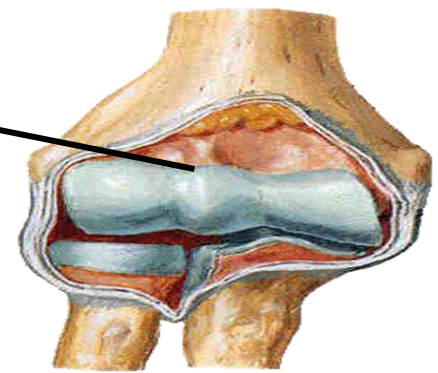
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The capsule is cut

Synovial membrane

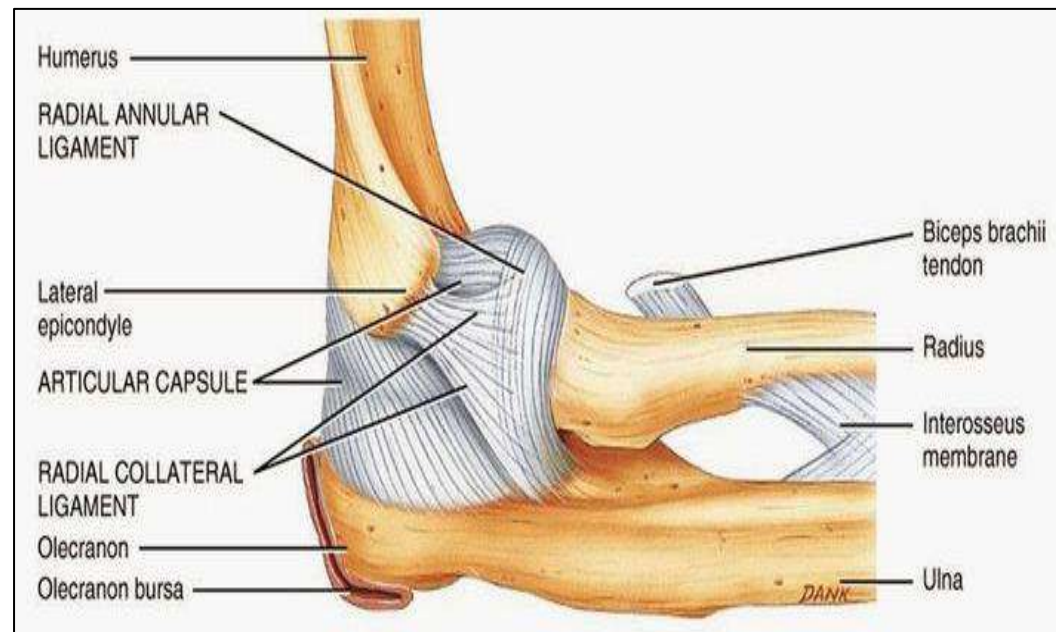
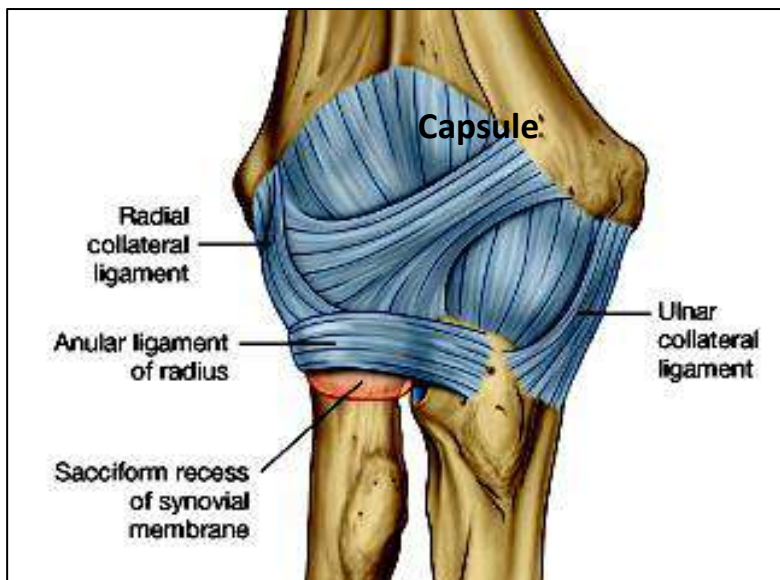
It lines the internal surface of the fibrous capsule



Lateral & Medial collateral ligaments of elbow joint

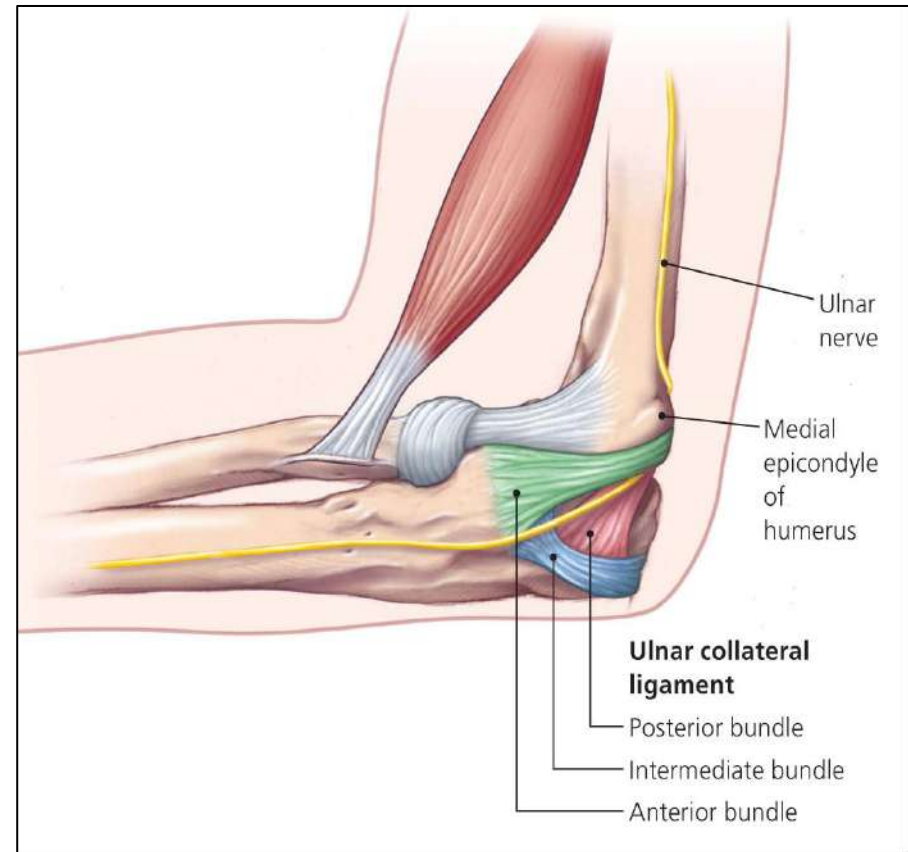
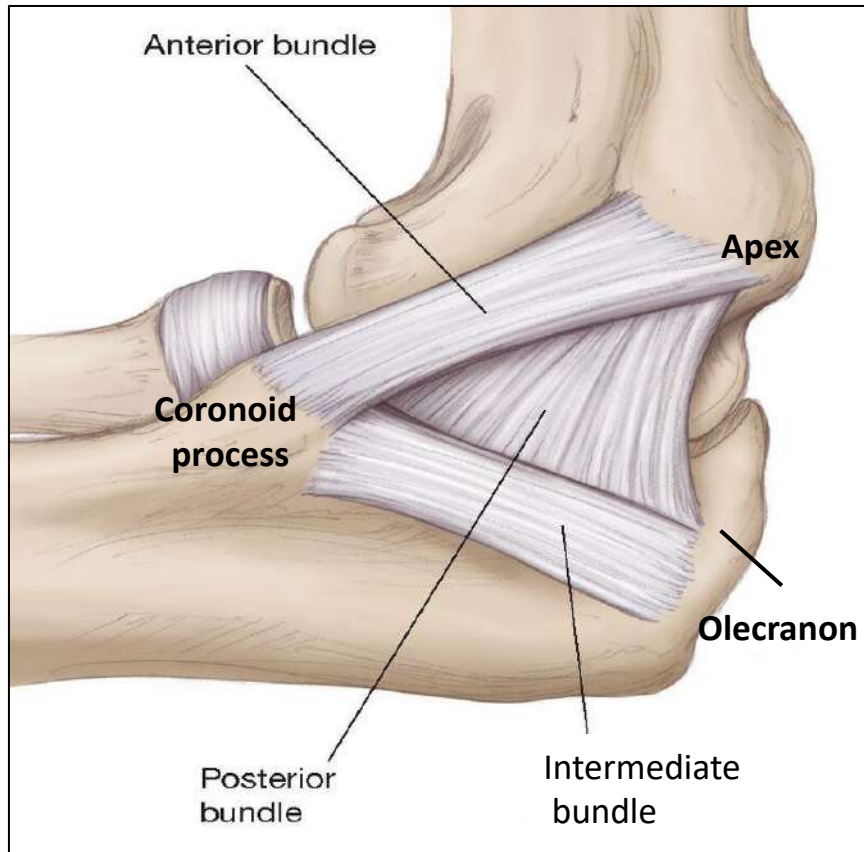
1. Lateral ligament (radial collateral)

Is triangular ligament, its apex is attached proximally to the lateral epicondyle of the humerus and its base blends with the annular ligament of the radius



2. Medial (ulnar collateral) ligament

- It is composed of 3 bundles : anterior & posterior bands which are connected by a thinner relatively weak **oblique (intermediate) bundle**.
 - Its apex is attached to the medial epicondyle of humerus.
 - The strong cord-like **anterior bundle** is attached to tubercle on coronoid process of ulna and the weak fan-like **posterior bundle** is attached to medial edge of olecranon.
- The **ulnar nerve** passes posterior to the medial epicondyle and is closely applied to the ulnar collateral ligament.



Movements & muscles of the Elbow joint

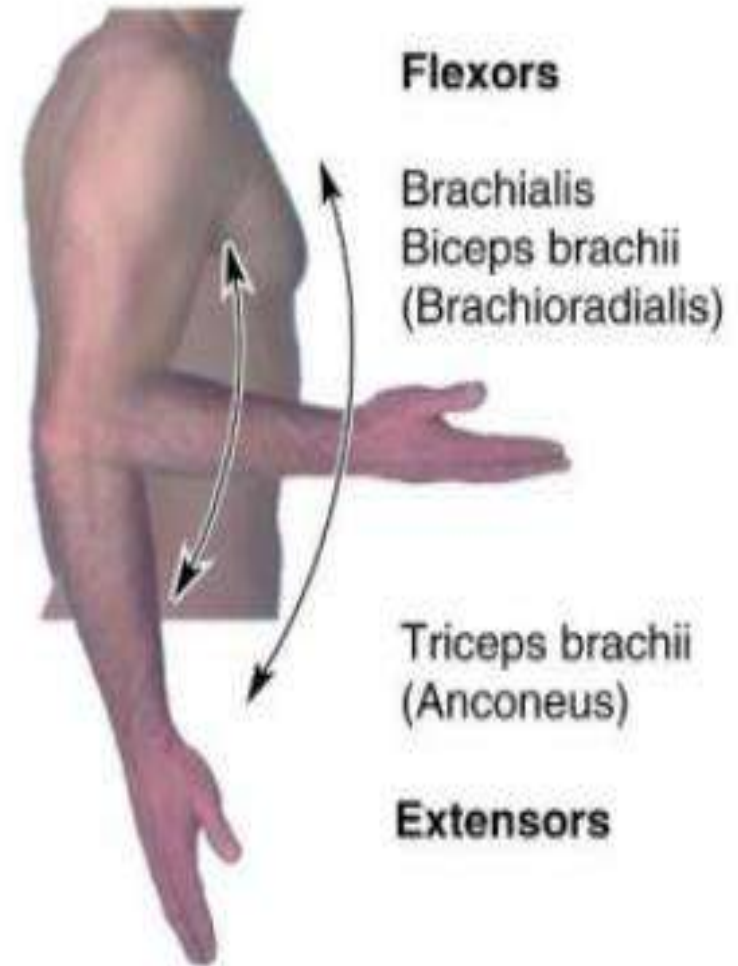
This joint can be flexed or extended , as it is moved around a transverse axis (uniaxial hinge joint)

Flexion is produced by :

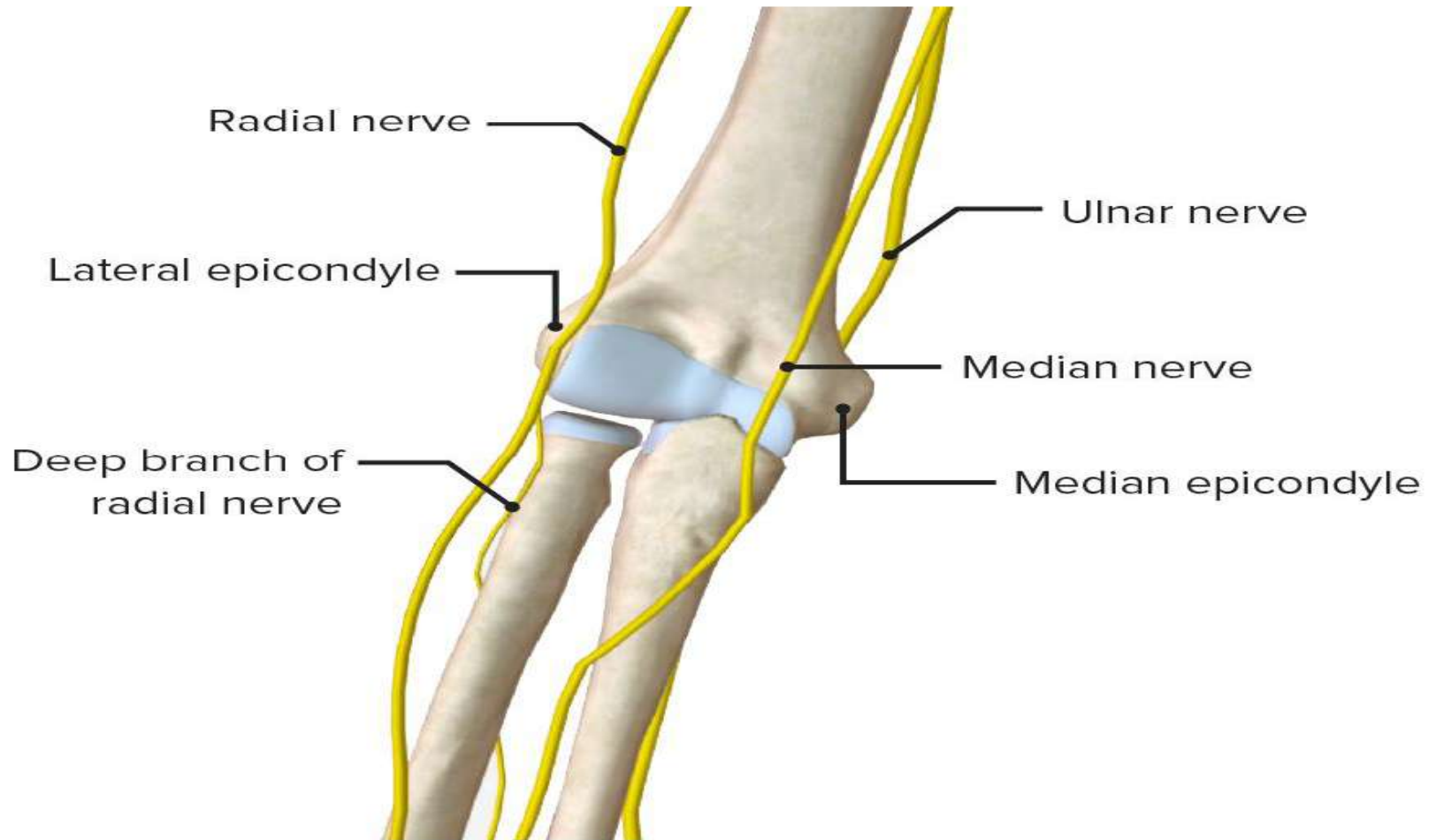
1. Brachialis is the main flexor
2. brachioradialis muscles
3. biceps brachii muscle also flexes this joint when the forearm is supinated
4. Pronator teres , also flexes the joint when the forearm is pronated

Extension is produced by

1. **Triceps muscle** , is the main extensor of the elbow joint .
2. **Anconeus** muscle assists with this movement.



Nerve supply of elbow joint

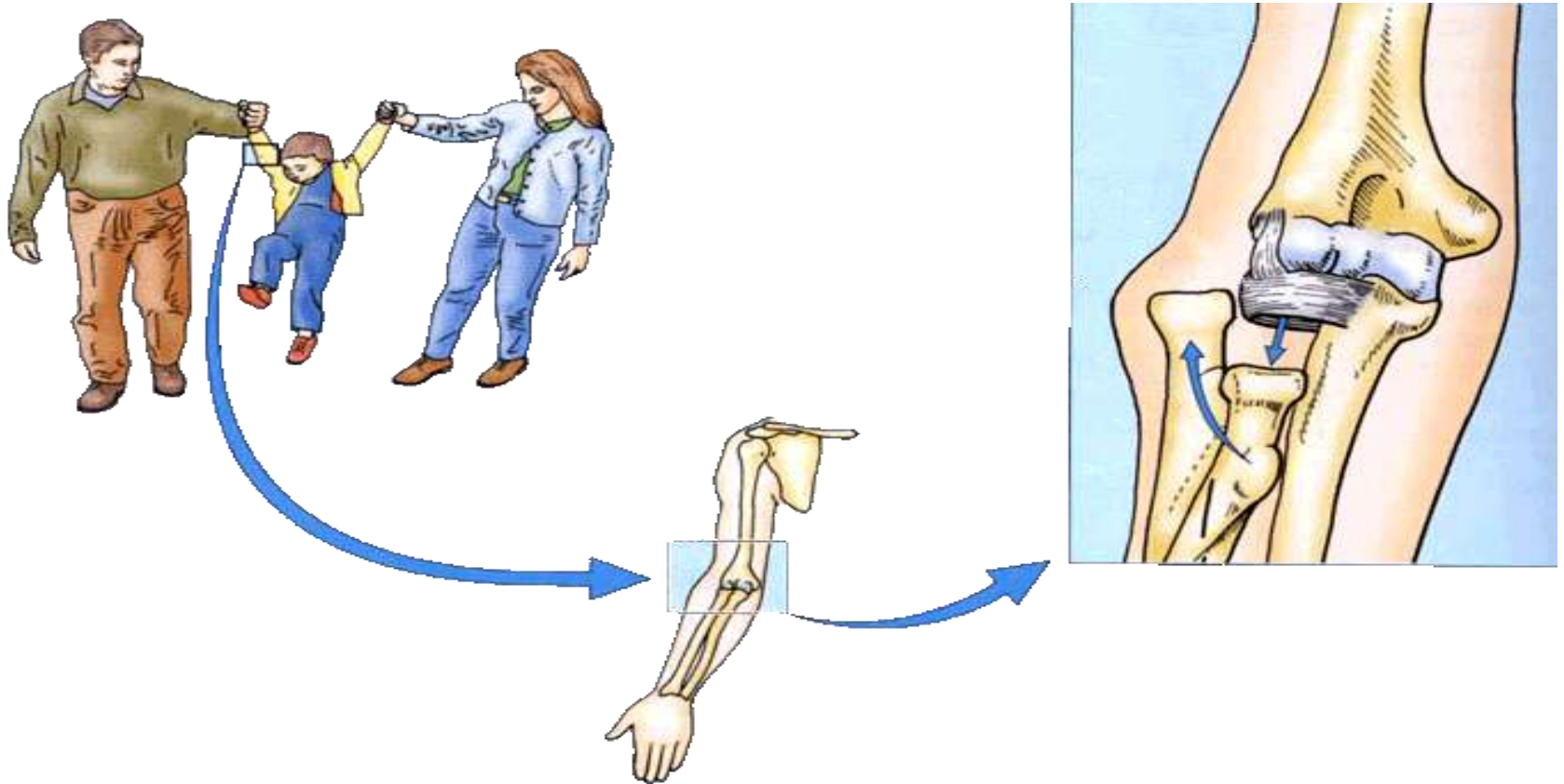


Dislocation of elbow joint

Elbow dislocations are common, and most are **posterior**.

Posterior dislocation usually follows falling on the outstretched hand.

Posterior dislocations of the joint **are common in children** because **the parts of the bones that stabilize the joint are incompletely developed**.



Superior (Proximal) Radioulnar Joint

Type : synovial pivot joint

Articular surfaces: the head of the radius (axis) and the annular ligament and the radial notch on the ulna (ring)

Capsule: it encloses the joint and is continuous with that of the elbow joint.

Ligament: the **annular ligament** is attached to the anterior and posterior margins of the radial notch of ulna and forms a collar around the head of the radius.

Synovial membrane: lines the capsule & is continuous above with that of the elbow joint.

Nerve supply: median, musculocutaneous, and radial nerves

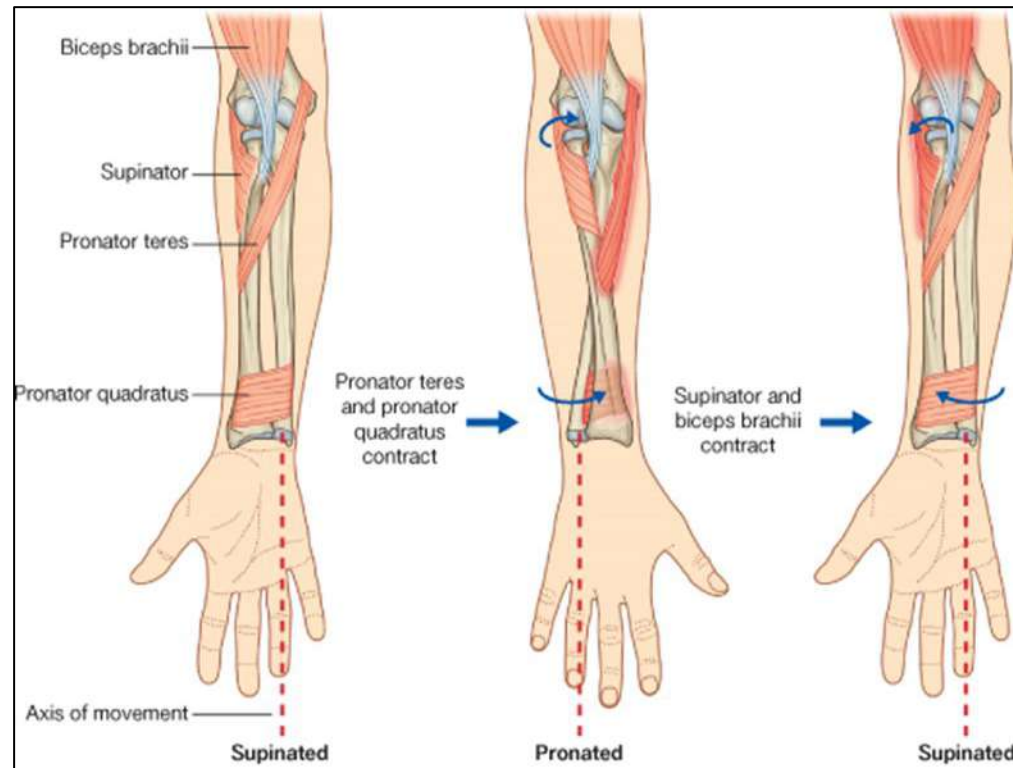
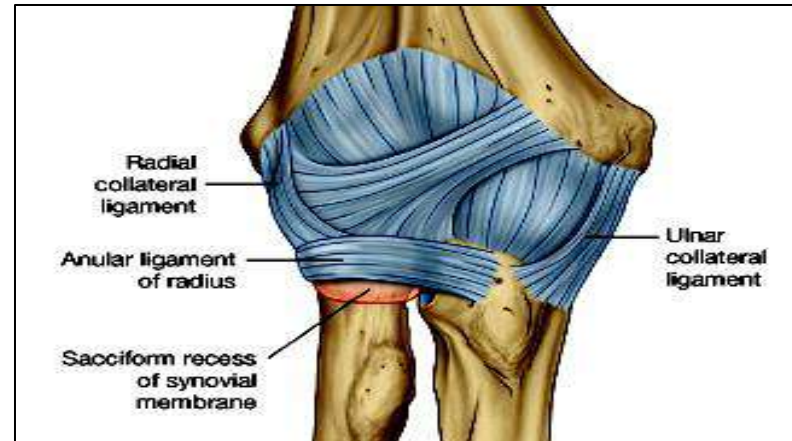
Movements : around a vertical axis .

Pronation & Supination .

Pronation: pronator teres and the pronator quadratus.

Supination: biceps brachii and the supinator.

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PA-Radiograph of extended elbow joint



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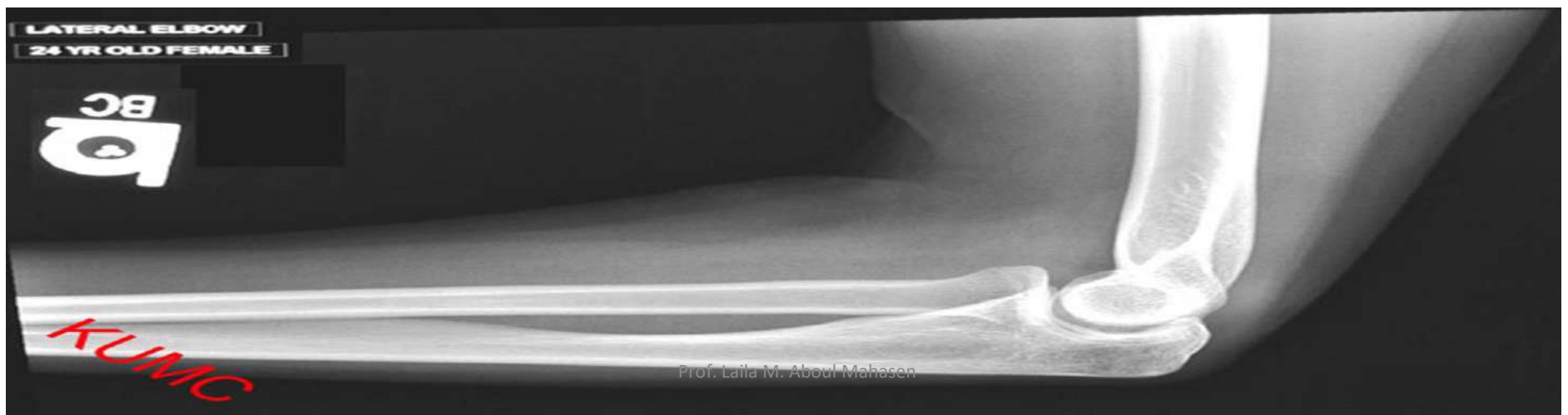
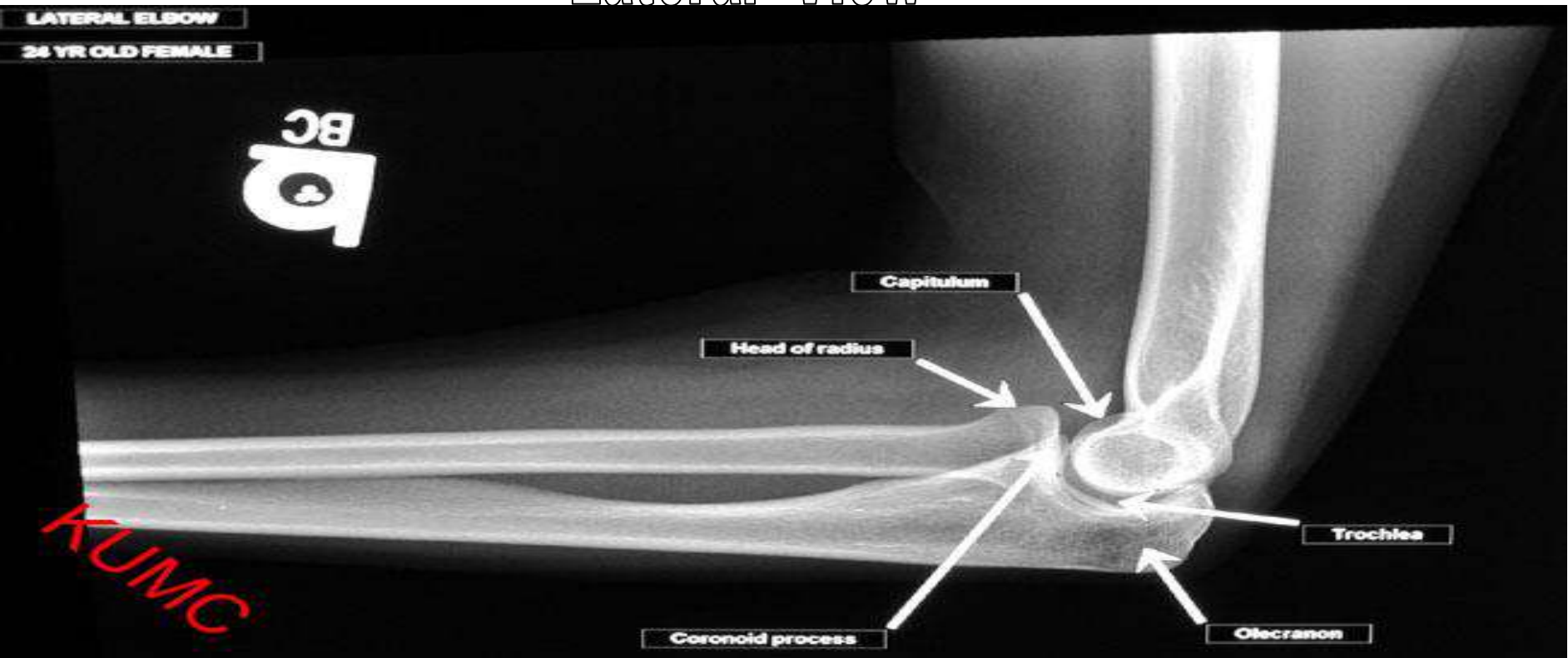
1. Medial humeral epicondyle 2. Shadow of olecranon superimposed on trochlea
3. Olecranon fossa 4. Lateral epicondyle 5. Capitulum
6. Humero-radial joint 7. Head of radius 8. Radial tuberosity
9. Radial head articulating with radial notch of ulna 10. Humero-ulnar joint

Lateral view Radiograph of semiflexed elbow joint

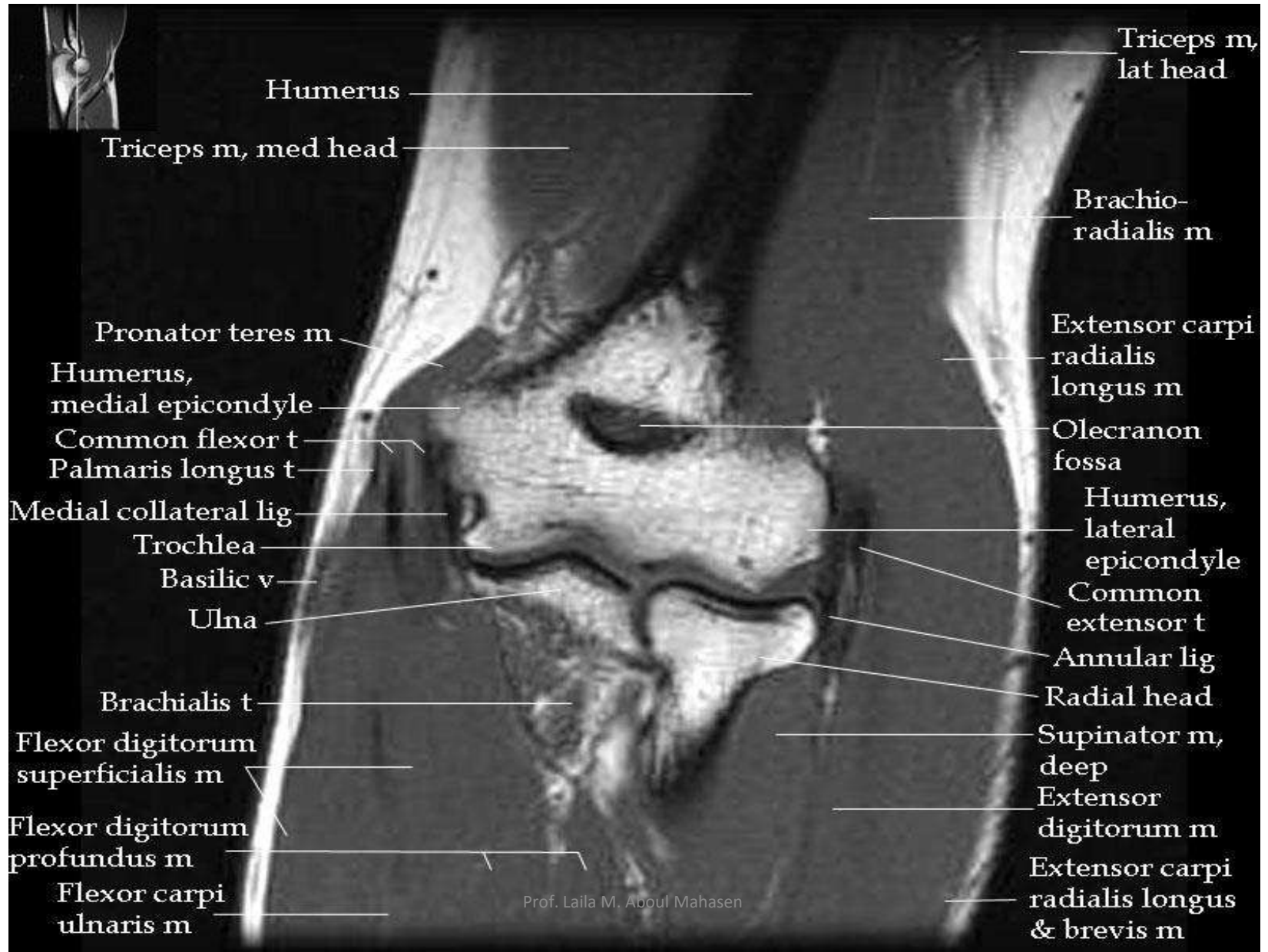


1. Head of radius 2. Profile of capitulum 3. Profile of trochlea 4. Olecranon

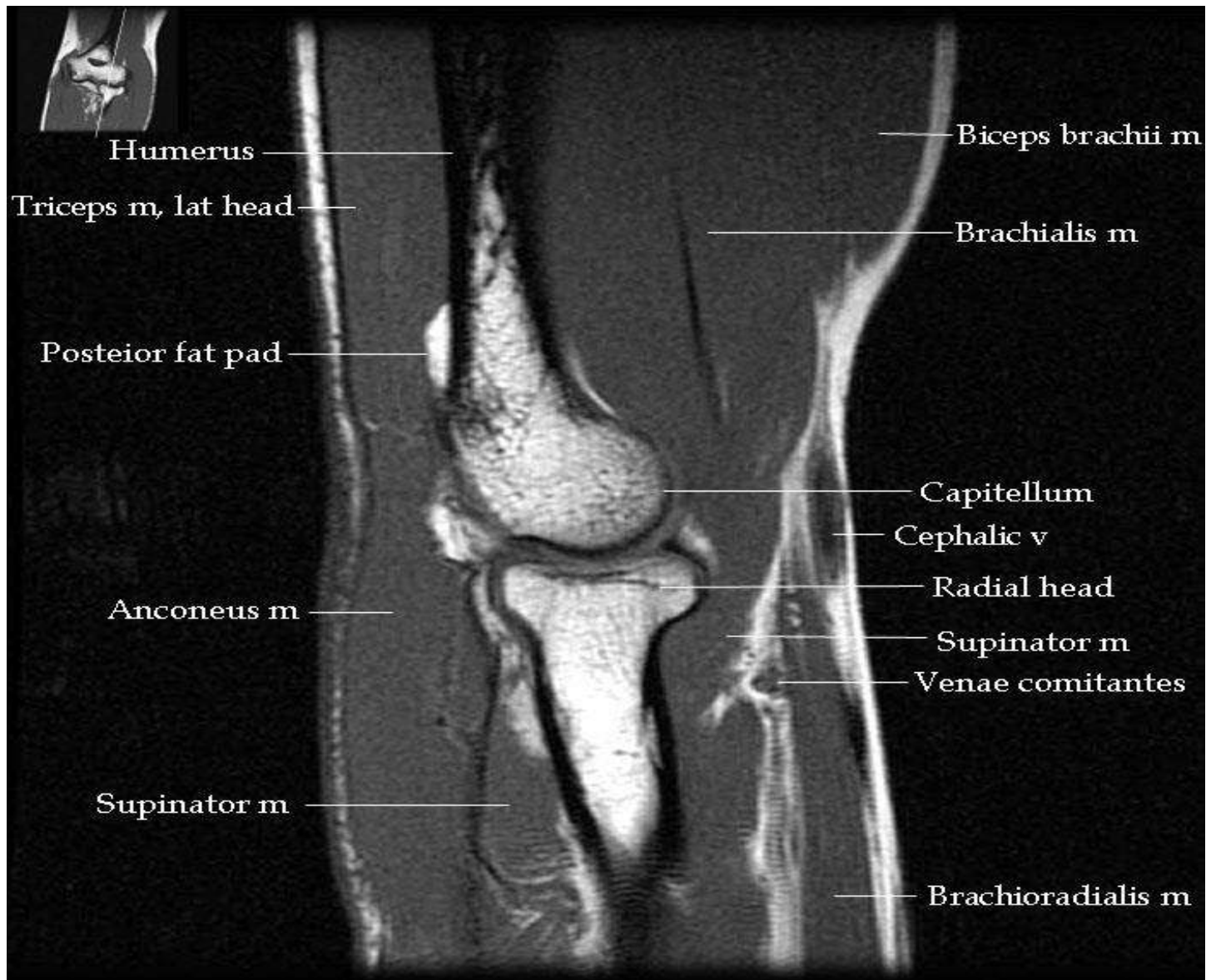
Lateral View



MRI Image



MRI Image



Inferior (Distal) Radioulnar Joint

Type : Synovial pivot joint

Articular surfaces: the head of the ulna and the ulnar notch of the radius

Capsule: it encloses the joint.

Ligaments: weak anterior and posterior ligaments strengthen the capsule.

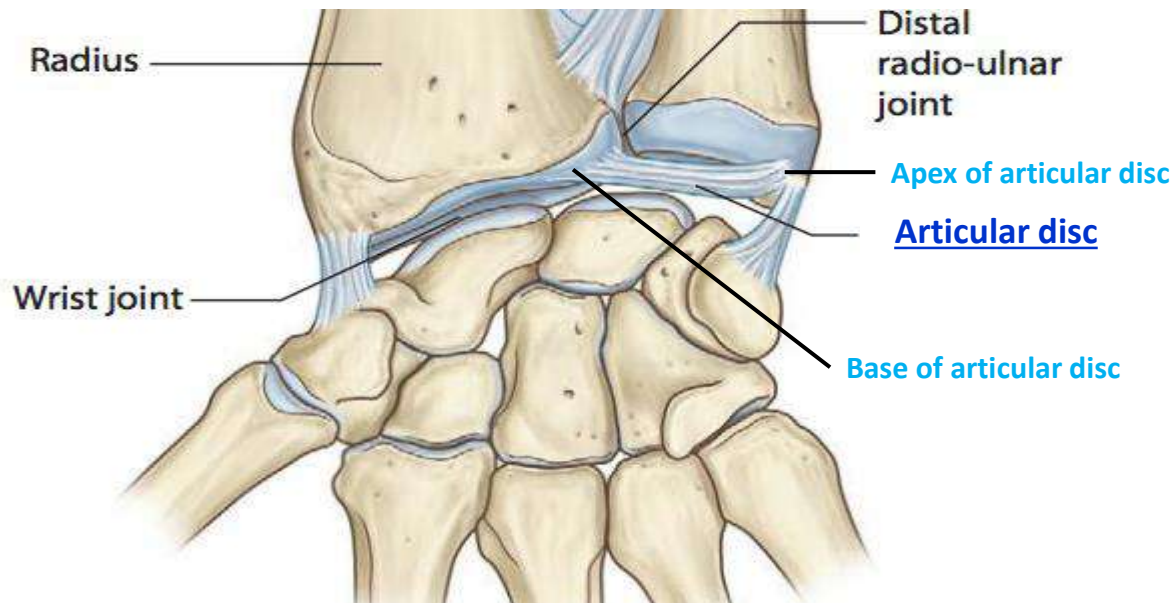
Articular disc: it is triangular fibrocartilage disc. Attached by its **apex** to lateral side of base of styloid process of ulna and by its **base** to lower border of ulnar notch of radius.

Synovial membrane: It lines the capsule.

Nerve supply: Anterior & posterior interosseous nerves

Movements of radioulnar joints:

- **Pronation:** pronator teres & pronator quadratus.
- **Supination:** biceps brachii & supinator.



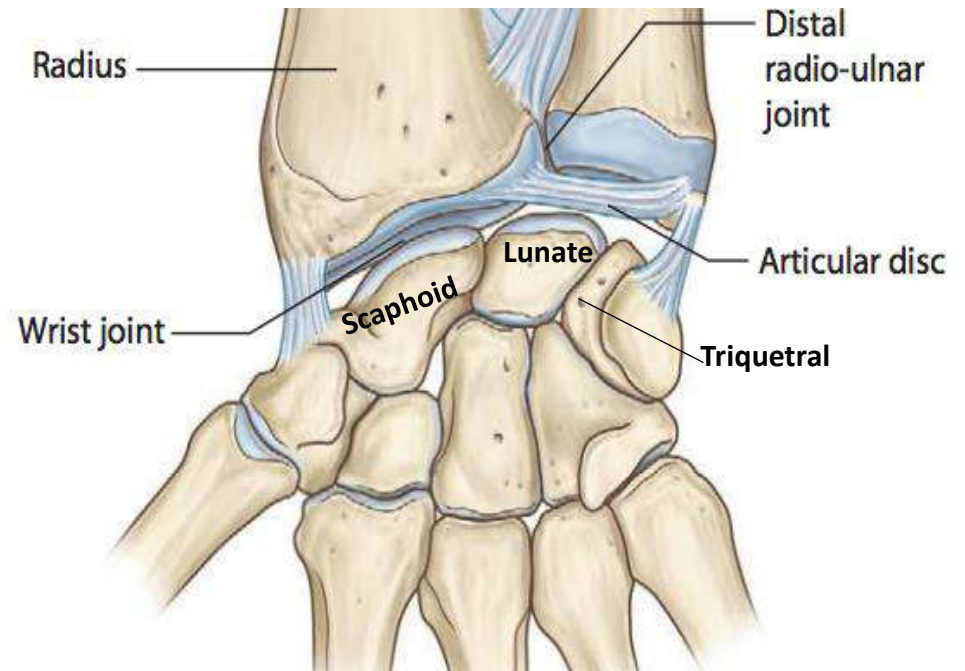
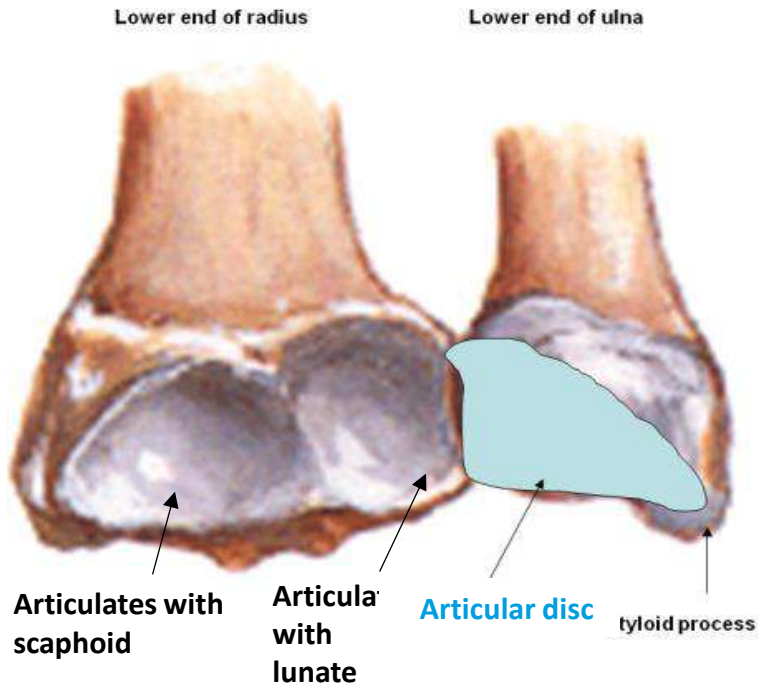
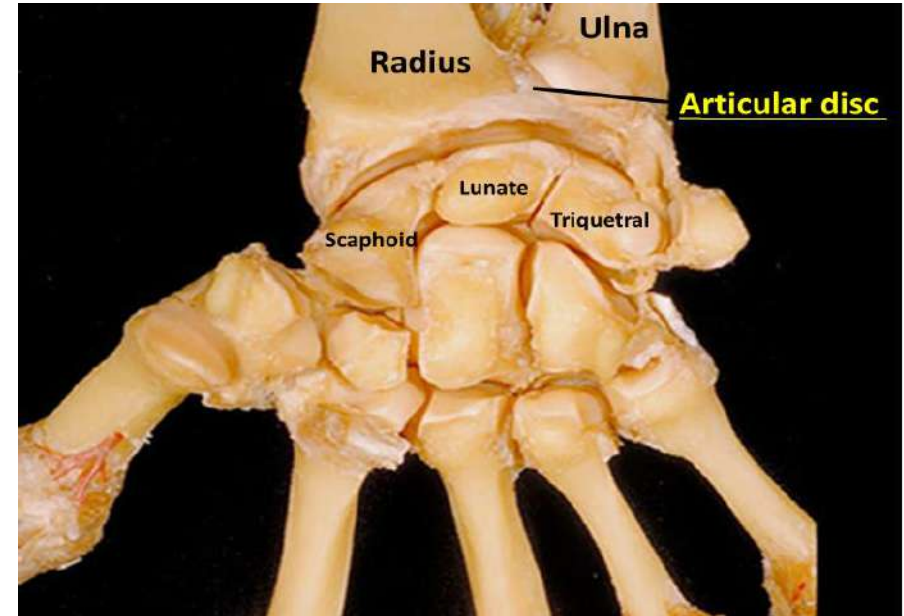
The Wrist joint (radiocarpal joint)

Type : synovial **condyloid** joint (biaxial)

Articular Surfaces :

Proximal : distal end of the radius and the **articular disc**.

Distal : scaphoid, lunate & triquetral



Capsule of Wrist joint

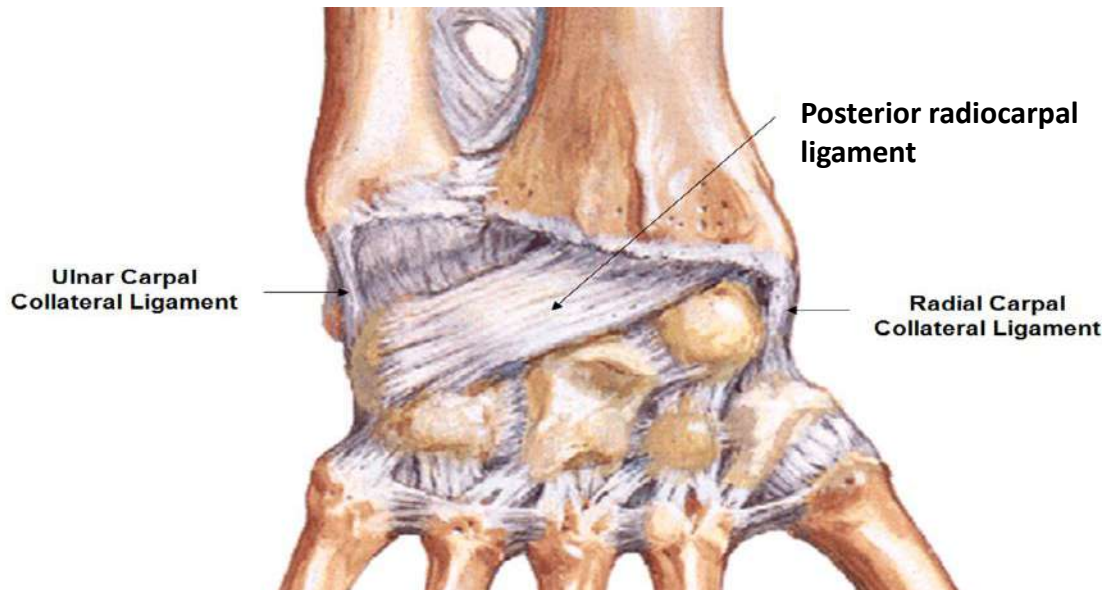
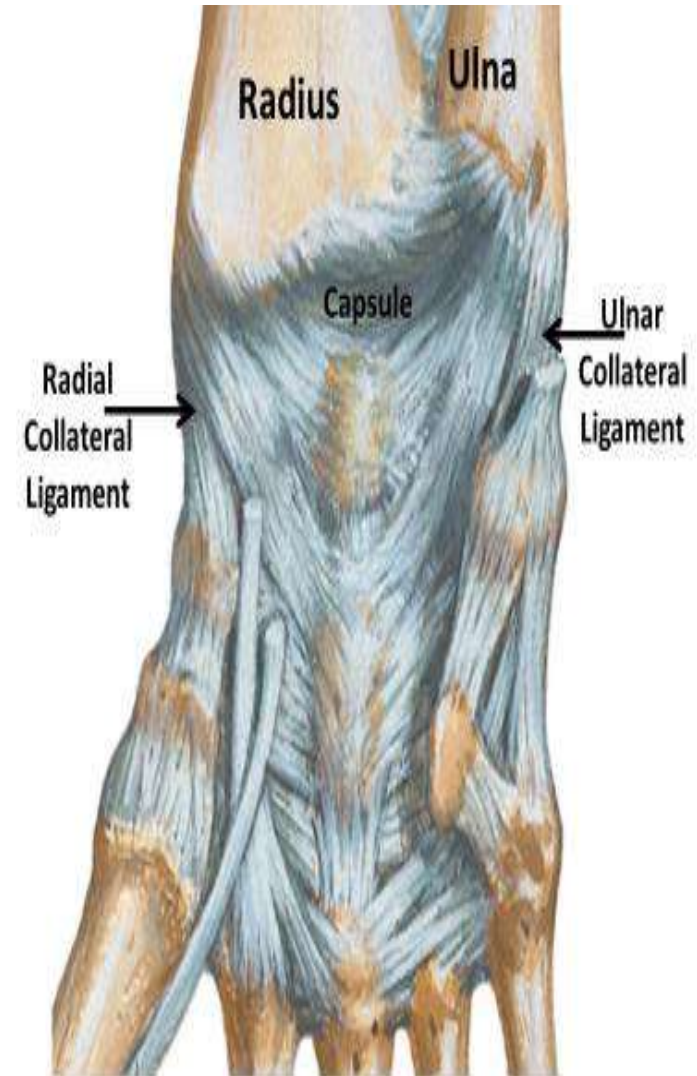
The capsule encloses the joint & is attached above to the distal ends of radius & ulna below to the proximal row of carpal bones

Ligaments of Wrist Joint

1& 2. Anterior & posterior radiocarpal ligaments strengthen the capsule.

3. Medial (Ulnar) collateral ligament is attached to styloid process of ulna & triquetral bone.

4. Lateral (Radial) collateral ligament is attached to styloid process of radius & scaphoid bone

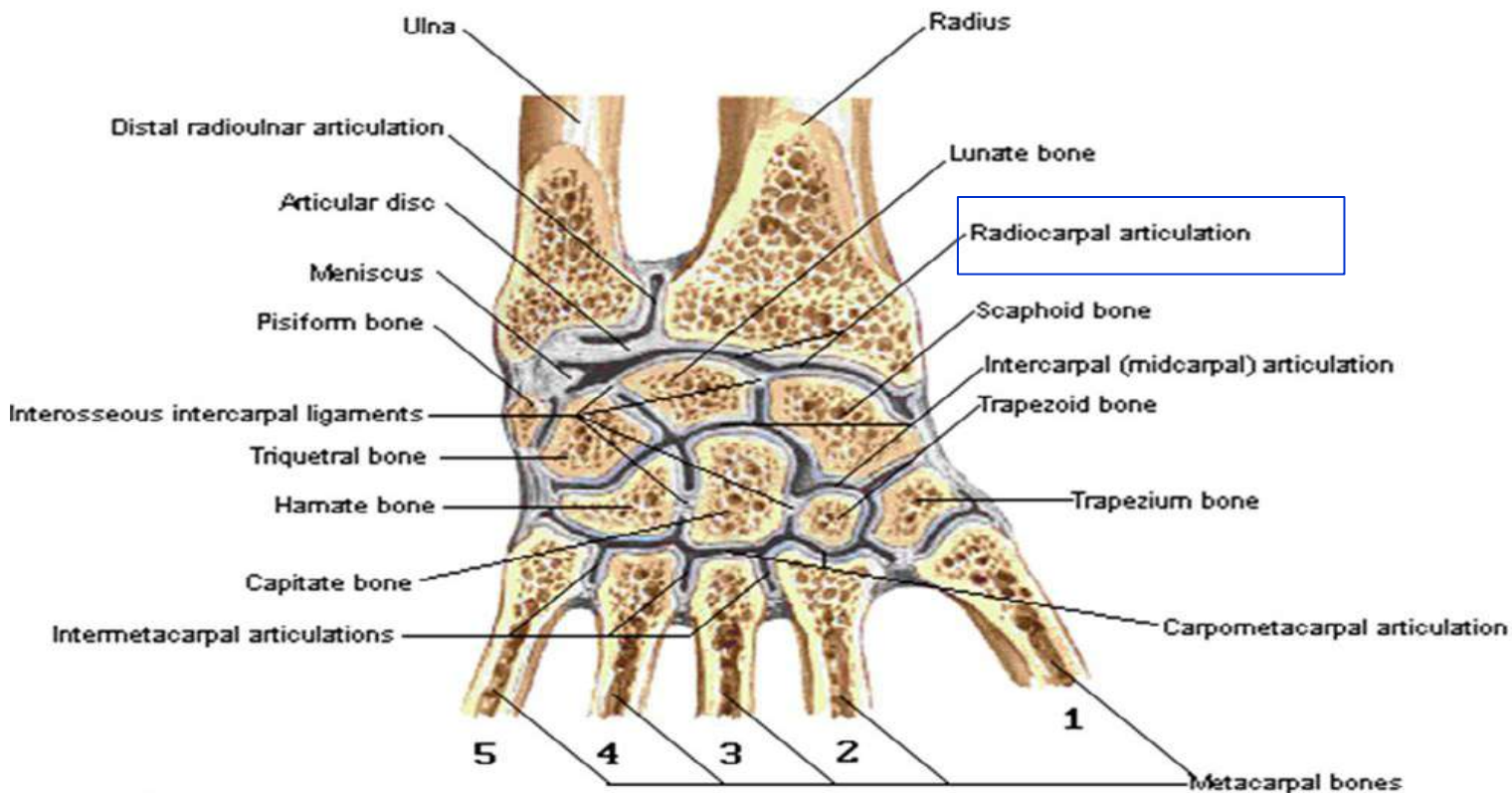


Synovial membrane

It lines the capsule & attached to the margins of the articular surfaces.

The joint cavity does not communicate with that of distal radioulnar joint or with the joint cavities of intercarpal joints

Coronal Section - Dorsal View

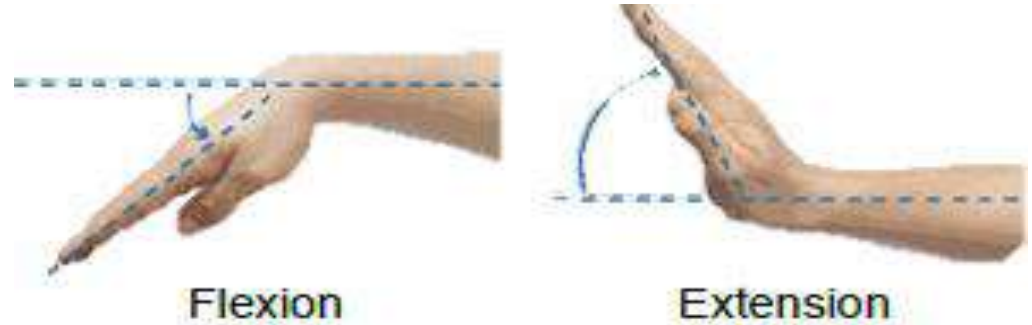


Movements of the WRIST JOINT

Movements : (Biaxial)

A. Around a transverse axis :

1. **Flexion** : all superficial & deep flexors of forearm
2. **Extension** : all superficial & deep extensors of forearm



B. Around a vertical axis :

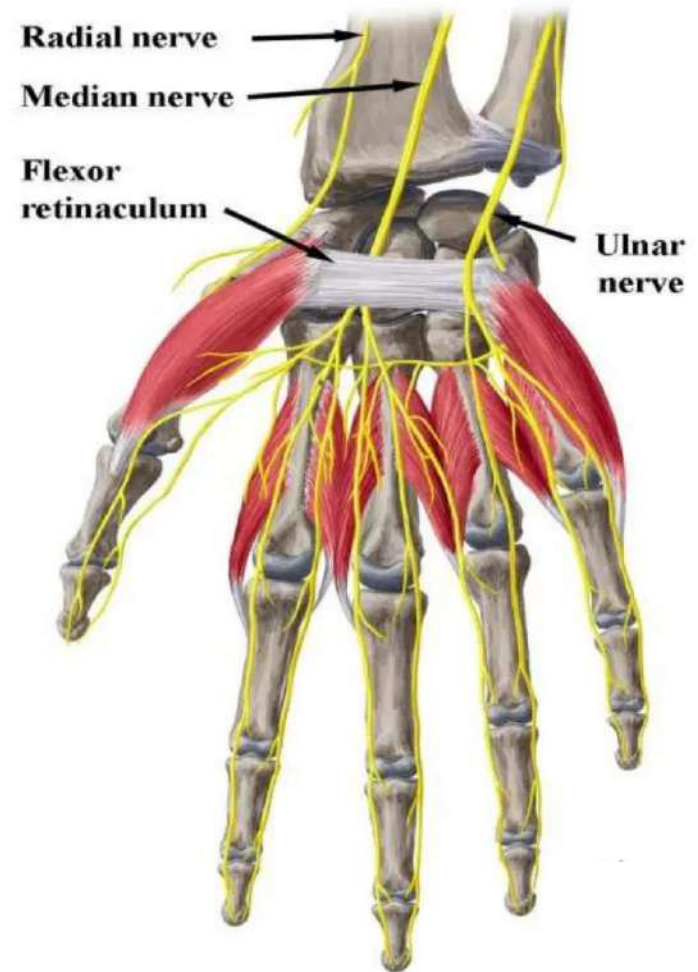
1. **Adduction (ulnar deviation)**: flexor & extensor carpi ulnaris
2. **Abduction (radial deviation)** : flexor & extensor carpi radialis



Nerve supply of wrist joint

By articular branches from the three nerves:

1. **Median nerve** → Anterior interosseous branch.
2. **Radial nerve** → Posterior interosseous branch.
3. **Ulnar nerve** → Deep and dorsal branches.



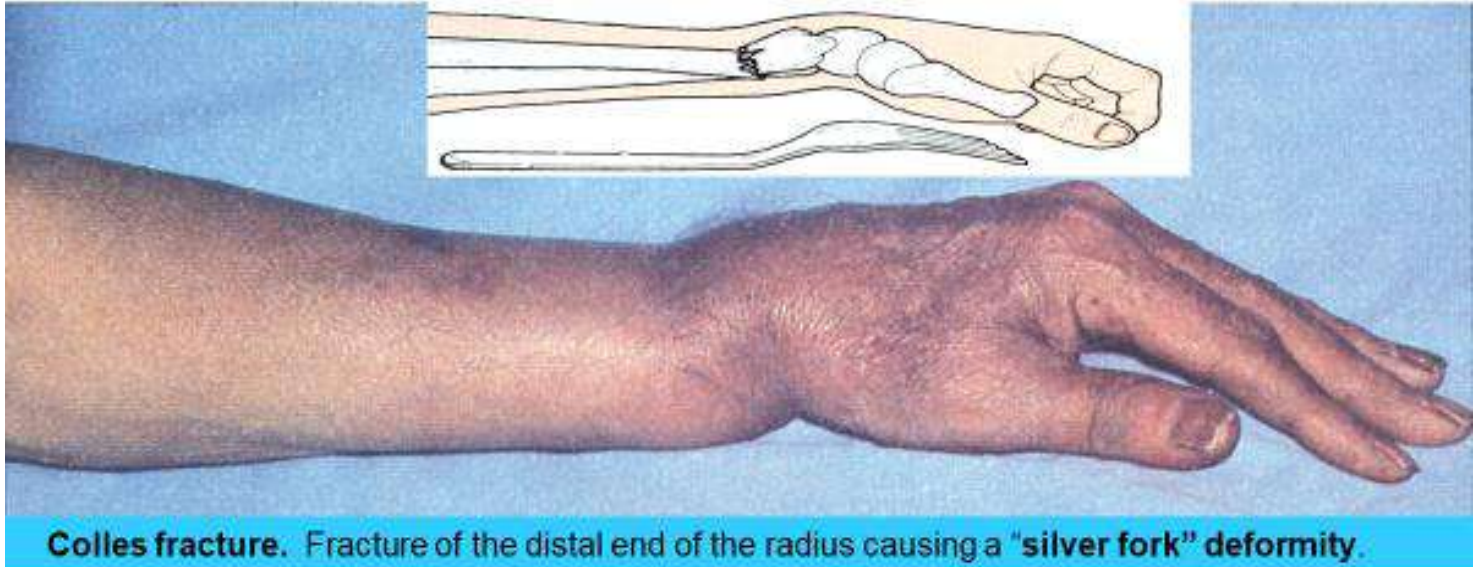
Wrist joint injury

A fall on the outstretched hand can strain the **anterior ligament of the wrist joint**, producing synovial effusion, joint pain, and limitation of movement. These symptoms and signs must not be confused with those produced by a fractured scaphoid or dislocation of the lunate bone, which are similar.

***If the forces are excessive**, different parts of the upper limb give way under the strain.

The area affected seems to be related to age.

1. **In a young child**, for example, there may be a posterior displacement of the distal radial epiphysis;
2. **in the teenager**, the clavicle might fracture;
3. **in the young adult**, the scaphoid is commonly fractured;
4. **in the elderly**, the distal end of the radius is fractured about 1 in. (2.5 cm) proximal to the wrist joint (**Colle's fracture**).



Colles fracture. Fracture of the distal end of the radius causing a "silver fork" deformity.



Distal
Interphalangeal
Joint

Proximal
Interphalangeal
Joint

Metacarpo-
phalangeal
Joint

Hamate
Capitate
Triquetrum
Pisiform
Lunate
Ulna Styloid
Process

Ulna

Distal Phalanx

Middle Phalanx

Phalanges

Proximal Phalanx

Metacarpals

Trapezoid
Trapezium

Scaphoid **Carpals**

Radial Styloid
Process

Radius



Distal
Interphalangeal
Joint

Proximal
Interphalangeal
Joint

Metacarpo-
phalangeal
Joint

Distal Phalanx

Middle Phalanx

Phalanges

Proximal Phalanx

Metacarpals

Hamate
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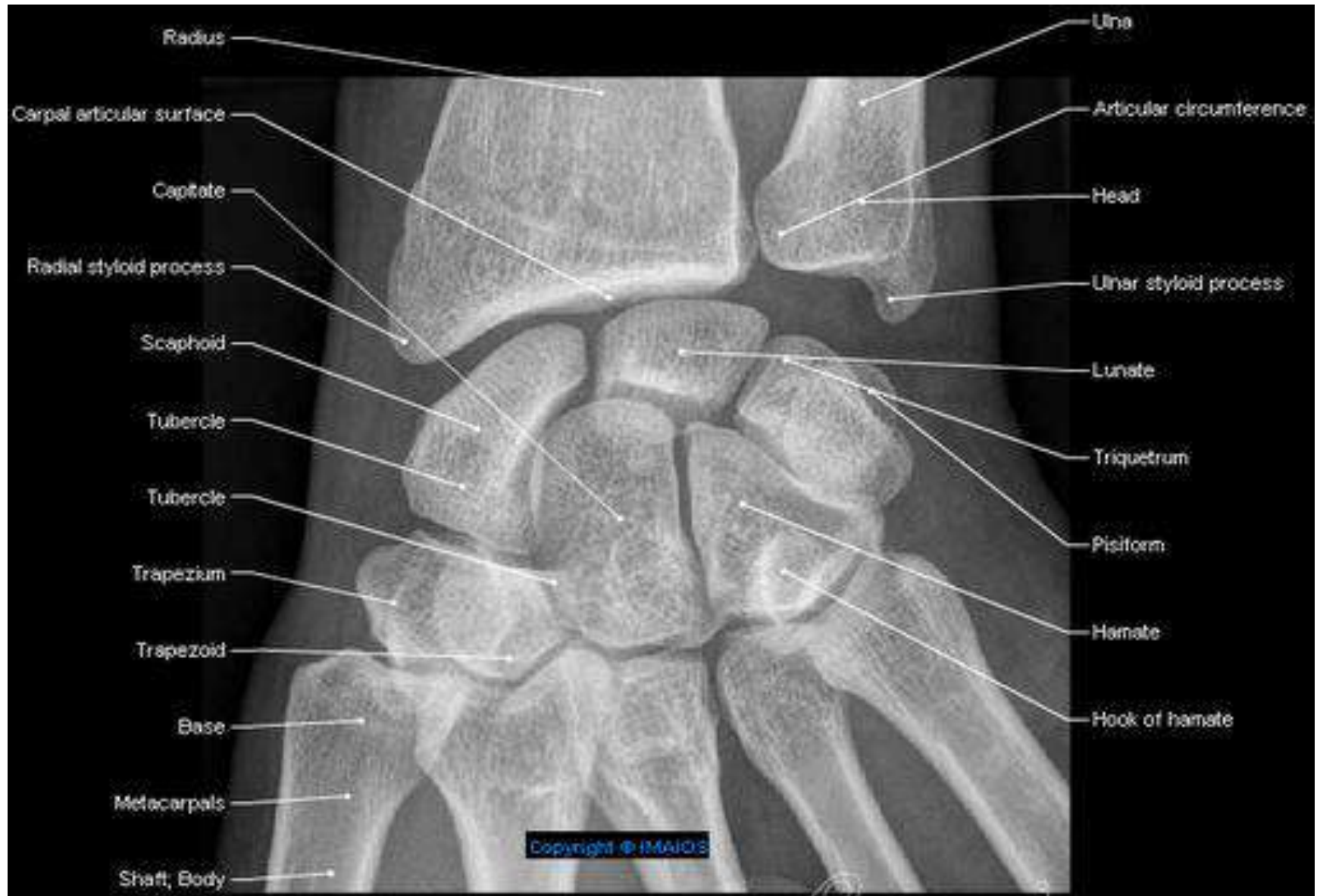
Scaphoid

Radial Styloid
Process

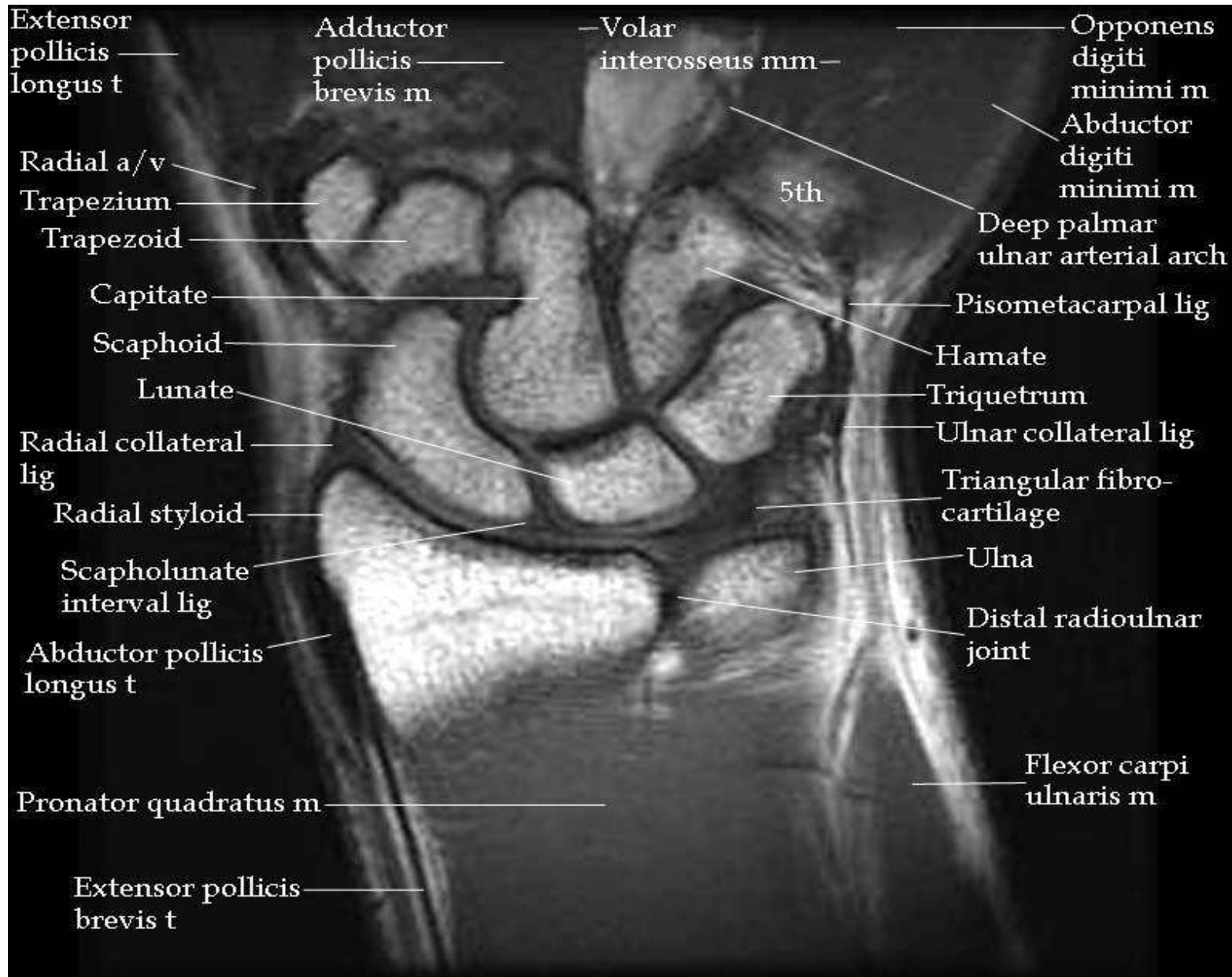
Carpals

Radius

Posterior view Radiograph



MRI –image



الحمد لله
على ما تعاقب الليل والنهار